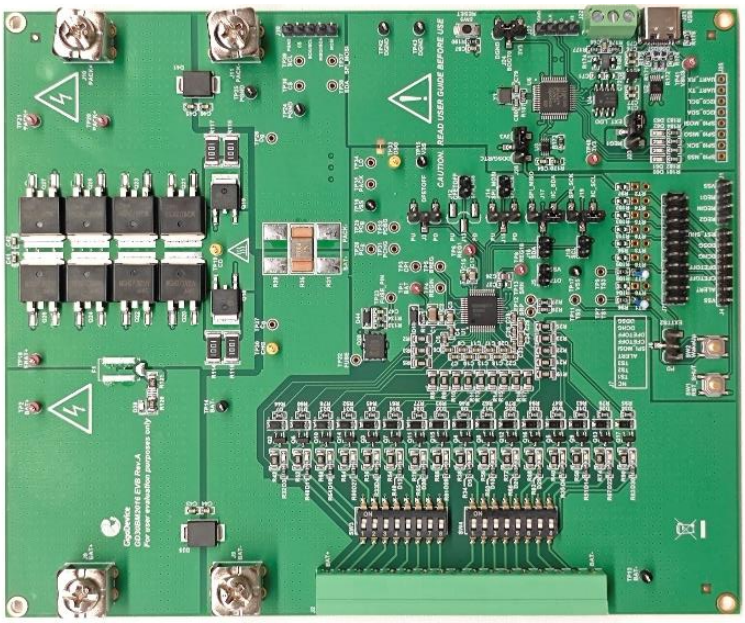
GD30BM2016
BMS Test Report

GD30BM2016EVB 是用于评估 GD30BM2016 3~16 串高精度电池管理系统监测与保护芯片的完整解决方案。本报告提供系统主要性能测试结果，可供电池管理开发者了解 GD30BM2016 性能及系统设计参考。



GD30BM2016 Evaluation Board

特性

- 精确监测3-16串电池组的单体电压、电流和温度
- 集成高度可配置的保护子系统
- 支持奇偶均衡，最多支持8串均衡功能
- 提供过充和过放保护，有效增强电池安全性和性能
- 支持400kHz I2C和16MHz SPI协议通信
- 支持多种低功耗运行模式

应用场景

- 便携式电源解决方案
- 低压家庭储能应用
- 锂电池驱动的两轮车系统
- 高压电动工具供电
- 用于各类工业电池组的应用场景

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目录

1 概述 3
2 GD30BM2016 EVB 设计文件 4
3 测试设备 5
4 GD30BM2016 EVB 主要规格及测试结果汇总 6
5 特性报告 7
5.1 系统电流消耗 7
5.2 电芯电压采样精度 7
5.3 TOS、LD、PACK 电压采样精度 11
5.4 电池组电流测量精度 13
5.5 温度测量精度 15
5.6 REG 输出 16
5.7 CP 输出 23
5.8 驱动 24
5.9 预放电转换 28
5.10 负载检测功能 29
5.11 容性负载启动 29
5.12 工作模式切换 30
5.13 电芯均衡 32
5.14 一次保护 34
6 热插拔 38
7 ESD 性能 39
8 历史版本 40

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1 概述

GD30BM2016EVB是用于评估GD30BM2016 3~16 串高精度电池管理系统监测与保护芯片的完整解决方案。本报告对EVB主要性能进行测试评估，评估内容包括电芯电压采样精度、电流采样精度、温度采样精度、LDO、驱动、电池均衡、电芯过压保护、电芯欠压保护、放电过流保护、充电过流保护、放电短路保护等。

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2 GD30BM2016 EVB 设计文件

请参阅 **GD30BM2016 EVB 用户指南。**

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3 测试设备

设备	制造商	型号
多通道模拟电池	NGI	N9124-06-01
数字示波器	Keysight	MSOX3104G
电子负载	Chorma	63630-80-60
数字万用表	Keysight	34470A
直流电源	Keysight	E36234A
源表	Keysight	B2962A
EVB	Gigadevice	GD30BM2016 EVB
上位机	Gigadevice	Battery Management System GUI
电池包	Gigadevice	30Ah

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4 GD30BM2016 EVB 主要规格及测试结果汇总

参数	条件	规格范围			测试结果			单位
		最小值	典型值	最大值	最小值	典型值	最大值	
电芯串数		3		16	3		16	串
均衡电流						83		mA
充电电流				25			25	A
放电电流				25			25	A
电芯电压精度				±5	-4		3	mV
电流精度	<2A						2	mA
	>2A						0.13	%
一次 OV 保护阈值			4200		4202		4204	mV
一次 OV 保护延迟时间			1			1		s
一次 UV 保护阈值			2700		2695		2700	mV
一次 UV 保护延迟时间			1			1		s
一级 OCD 保护阈值			25			25.4		A
一级 OCD 保护延迟时间			1			1		s
二级 OCD 保护阈值			50			50.7		A
二级 OCD 保护延迟时间			0.5			0.5		s
OCC 保护阈值			12			12		A
OCC 保护延迟时间			0.5			0.5		s
SCD 保护阈值			100			101		A
SCD 保护延迟时间			60			60		us
充电 OTP			55			54		℃
放电 OTP			60			59		℃
充电 UTP			0			-0.8		℃
放电 UTP			-20			-21		℃
总关断电流			5			4.5		uA

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5 特性报告

5.1 系统电流消耗

Tab. 5-1-1 系统电流消耗

模式	IBAT(uA)						
	VBAT=8V	VBAT=16V	VBAT=32V	VBAT=48V	VBAT=59.2V	VBAT=67.2V	VBAT=80V
正常模式（通信）	1875	1877	1877	1877	1877	1877	1877
正常模式（不通信）	1046	1047	1047	1049	1049	1052	1052
睡眠模式	52	53	54	54	54	54	54
深度睡眠模式	25.2	25.3	25.6	25.6	25.8	25.8	26.2
关断模式	3.9	4.1	4.3	4.4	4.5	4.6	4.9

5.2 电芯电压采样精度

Tab.5-2-1 电芯采样电压@25℃

实测电压	Cell Voltage(mV)															
(mV)	Cell1	Cell2	Cell3	Cell4	Cell5	Cell6	Cell7	Cell8	Cell9	Cell10	Cell11	Cell12	Cell13	Cell14	Cell15	Cell16
1500	1502	1501	1502	1501	1501	1500	1501	1501	1501	1501	1501	1501	1501	1501	1500	1501
1750	1751	1751	1751	1751	1751	1750	1751	1751	1751	1751	1751	1751	1751	1751	1751	1750
2000	2001	2001	2001	2001	2000	2001	2001	2001	2001	2001	2001	2001	2001	2001	2002	2000
2250	2251	2251	2251	2250	2251	2251	2251	2251	2252	2250	2251	2251	2251	2251	2251	2251
2500	2501	2501	2500	2501	2501	2501	2501	2501	2500	2501	2501	2501	2501	2501	2501	2500
2750	2751	2751	2750	2750	2751	2752	2751	2750	2751	2750	2751	2751	2751	2751	2751	2750
3000	3001	3000	3000	3001	3000	3001	3001	3001	3001	3001	3001	3000	3001	3001	3000	3000
3250	3251	3249	3250	3250	3251	3250	3250	3251	3250	3249	3250	3250	3250	3250	3250	3250
3500	3501	3500	3501	3500	3500	3500	3501	3500	3500	3500	3500	3500	3501	3500	3500	3499
3750	3751	3750	3750	3750	3750	3751	3750	3750	3750	3751	3750	3750	3750	3750	3750	3750
4000	4001	4001	4000	4000	4000	4000	4001	4000	4001	4000	4000	4001	4000	4000	4000	3999
4250	4250	4250	4250	4250	4249	4250	4250	4249	4250	4250	4250	4250	4250	4250	4249	4249
4500	4500	4499	4500	4499	4499	4499	4500	4499	4499	4499	4499	4499	4499	4500	4499	4499

Tab.5-2-2 电芯电压采样精度@25℃

实测电压	Cell Voltage Error(mV)															
(mV)	Cell1	Cell2	Cell3	Cell4	Cell5	Cell6	Cell7	Cell8	Cell9	Cell10	Cell11	Cell12	Cell13	Cell14	Cell15	Cell16
1500	2	1	2	1	1	0	1	1	1	1	1	1	1	1	0	1
1750	1	1	1	1	1	0	1	1	1	1	1	1	1	1	1	0
2000	1	1	1	1	0	1	1	1	1	1	1	1	1	1	2	0
2250	1	1	1	0	1	1	1	1	2	0	1	1	1	1	1	1
2500	1	1	0	1	1	1	1	1	0	1	1	1	1	1	1	0
2750	1	1	0	0	1	2	1	0	1	0	1	1	1	1	1	0
3000	1	0	0	1	0	1	1	1	1	1	1	0	1	1	0	0

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实测电压	Cell Voltage Error(mV)															
(mV)	Cell1	Cell2	Cell3	Cell4	Cell5	Cell6	Cell7	Cell8	Cell9	Cell10	Cell11	Cell12	Cell13	Cell14	Cell15	Cell16
3250	1	-1	0	0	1	0	0	1	0	-1	0	0	0	0	0	0
3500	1	0	1	0	0	0	1	0	0	0	0	0	1	0	0	-1
3750	1	0	0	0	0	1	0	0	0	1	0	0	0	0	0	0
4000	1	1	0	0	0	0	1	0	1	0	0	1	0	0	0	-1
4250	0	0	0	0	-1	0	0	-1	0	0	0	0	0	0	-1	-1
4500	0	-1	0	-1	-1	-1	0	-1	-1	-1	-1	-1	-1	0	-1	-1

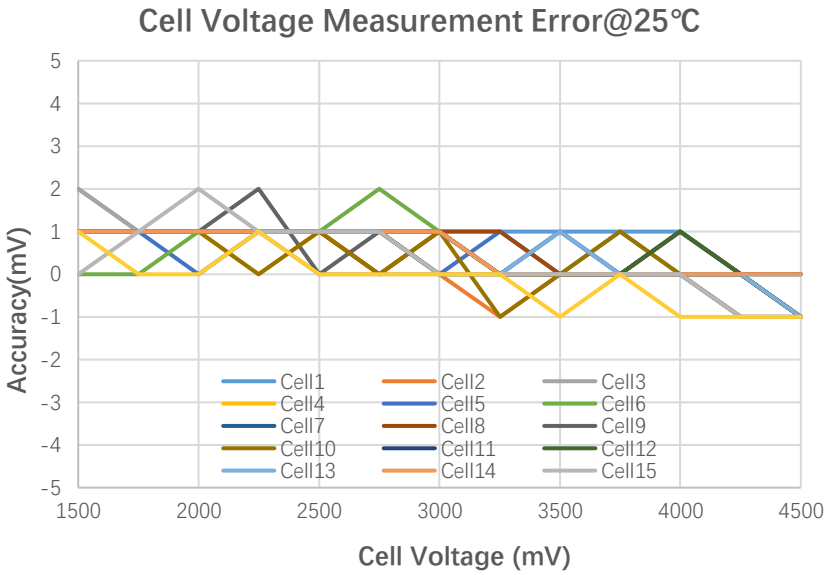


Fig. 5-2-1 电芯电压采样精度@25℃

Tab.5-2-3 电芯采样电压@-40℃

实测电压	Cell Voltage(mV)															
(mV)	Cell1	Cell2	Cell3	Cell4	Cell5	Cell6	Cell7	Cell8	Cell9	Cell10	Cell11	Cell12	Cell13	Cell14	Cell15	Cell16
1500	1503	1502	1503	1502	1502	1502	1502	1502	1502	1502	1502	1503	1502	1502	1502	1502
1750	1752	1752	1752	1752	1753	1752	1752	1752	1751	1752	1752	1752	1752	1752	1752	1751
2000	2003	2002	2003	2002	2002	2002	2002	2002	2002	2001	2002	2002	2002	2002	2002	2001
2250	2252	2252	2252	2251	2252	2252	2252	2252	2252	2252	2251	2252	2251	2252	2252	2251
2500	2502	2501	2501	2502	2502	2502	2502	2502	2502	2502	2502	2502	2502	2502	2502	2501
2750	2752	2751	2752	2752	2751	2751	2752	2751	2752	2752	2752	2752	2751	2752	2751	2750
3000	3002	3002	3002	3002	3002	3002	3002	3002	3002	3002	3002	3002	3001	3001	3002	3001
3250	3252	3251	3252	3252	3252	3252	3252	3252	3252	3251	3252	3251	3251	3251	3251	3250
3500	3502	3501	3502	3502	3502	3502	3502	3502	3502	3501	3502	3501	3501	3501	3501	3500
3750	3752	3752	3752	3752	3752	3752	3752	3752	3752	3751	3751	3751	3750	3750	3751	3751
4000	4002	4001	4002	4002	4002	4002	4002	4002	4002	4001	4002	4000	4001	4001	4001	4000
4250	4251	4251	4252	4251	4252	4251	4252	4252	4252	4251	4252	4250	4250	4251	4251	4250
4500	4501	4502	4502	4501	4501	4501	4501	4501	4501	4501	4501	4500	4500	4501	4501	4500

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Tab.5-2-4 电芯电压采样精度@-40℃

实测电压	Cell Voltage Error(mV)															
(mV)	Cell1	Cell2	Cell3	Cell4	Cell5	Cell6	Cell7	Cell8	Cell9	Cell10	Cell11	Cell12	Cell13	Cell14	Cell15	Cell16
1500	3	2	3	2	2	2	2	2	2	2	2	3	2	2	2	2
1750	2	2	2	2	3	2	2	2	1	2	2	2	2	2	2	1
2000	3	2	3	2	2	2	2	2	2	1	2	2	2	2	2	1
2250	2	2	2	1	2	2	2	2	2	2	1	2	1	2	2	1
2500	2	1	1	2	2	2	2	2	2	2	2	2	2	2	2	1
2750	2	1	2	2	1	1	2	1	2	2	2	2	1	2	1	0
3000	2	2	2	2	2	2	2	2	2	2	2	2	1	1	2	1
3250	2	1	2	2	2	2	2	2	2	1	2	1	1	1	1	0
3500	2	1	2	2	2	2	2	2	2	1	2	1	1	1	1	0
3750	2	2	2	2	2	2	2	2	2	1	1	1	0	0	1	1
4000	2	1	2	2	2	2	2	2	2	1	2	0	1	1	1	0
4250	1	1	2	1	2	1	2	2	2	1	2	0	0	1	1	0
4500	1	2	2	1	1	1	1	1	1	1	1	0	0	1	1	0

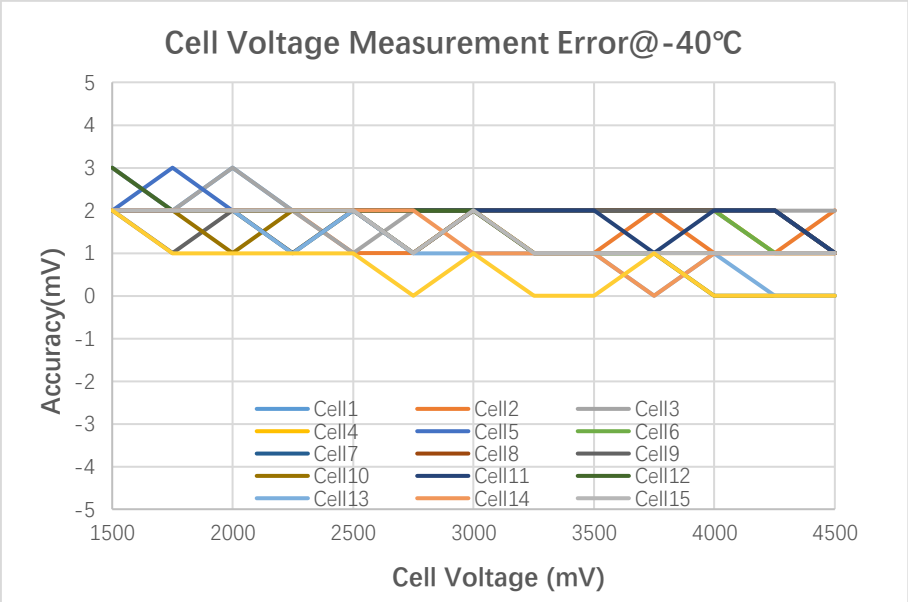


Fig. 5-2-2 电芯电压采样精度@-40℃

Tab.5-2-5 电芯采样电压@85℃

实测电压	Cell Voltage(mV)															
(mV)	Cell1	Cell2	Cell3	Cell4	Cell5	Cell6	Cell7	Cell8	Cell9	Cell10	Cell11	Cell12	Cell13	Cell14	Cell15	Cell16
1500	1500	1500	1499	1500	1500	1500	1500	1499	1500	1500	1500	1500	1499	1499	1500	1499
1750	1750	1749	1749	1750	1750	1750	1750	1750	1750	1749	1750	1749	1749	1749	1749	1748
2000	2000	1999	2000	1999	1999	1999	2000	2000	2000	1999	2000	1999	1999	1999	1999	1998
2250	2250	2249	2250	2249	2250	2250	2250	2249	2250	2249	2250	2249	2248	2249	2249	2248
2500	2499	2498	2499	2498	2500	2500	2500	2499	2499	2499	2499	2499	2499	2499	2499	2498

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2750	2749	2748	2749	2748	2750	2749	2749	2749	2749	2749	2749	2749	2748	2749	2748	2747
3000	2999	2999	2999	2999	2999	2999	2999	2999	2999	2999	2999	2999	2998	2998	2999	2998
3250	3250	3248	3250	3249	3248	3250	3249	3249	3249	3248	3250	3248	3248	3248	3248	3247
3500	3499	3498	3500	3499	3500	3499	3499	3499	3500	3498	3500	3498	3498	3498	3498	3497
3750	3749	3749	3750	3749	3749	3749	3749	3749	3749	3748	3749	3748	3747	3747	3748	3746
4000	3999	3998	3999	3999	3999	3999	4000	3999	4000	3998	3999	3997	3998	3998	3998	3997
4250	4248	4248	4249	4249	4249	4248	4249	4249	4249	4248	4249	4247	4247	4248	4248	4247
4500	4498	4499	4500	4499	4499	4499	4499	4499	4499	4498	4499	4497	4497	4497	4498	4496

Tab.5-2-6 电芯电压采样精度@85℃

实测电压	Cell Voltage Error(mV)															
(mV)	Cell1	Cell2	Cell3	Cell4	Cell5	Cell6	Cell7	Cell8	Cell9	Cell10	Cell11	Cell12	Cell13	Cell14	Cell15	Cell16
1500	0	0	-1	0	0	0	0	-1	0	0	0	0	-1	-1	0	-1
1750	0	-1	-1	0	0	0	0	0	0	-1	0	-1	-1	-1	-1	-2
2000	0	-1	0	-1	-1	-1	0	0	0	-1	0	-1	-1	-1	-1	-2
2250	0	-1	0	-1	0	0	0	-1	0	-1	0	-1	-2	-1	-1	-2
2500	-1	-2	-1	-2	0	0	0	-1	-1	-1	-1	-1	-1	-1	-1	-2
2750	-1	-2	-1	-2	0	-1	-1	-1	-1	-1	-1	-1	-2	-1	-2	-3
3000	-1	-1	-1	-1	-1	-1	-1	-1	-1	-1	-1	-1	-2	-2	-1	-2
3250	0	-2	0	-1	-2	0	-1	-1	-1	-2	0	-2	-2	-2	-2	-3
3500	-1	-2	0	-1	0	-1	-1	-1	0	-2	0	-2	-2	-2	-2	-3
3750	-1	-1	0	-1	-1	-1	-1	-1	-1	-2	-1	-2	-3	-3	-2	-4
4000	-1	-2	-1	-1	-1	-1	0	-1	0	-2	-1	-3	-2	-2	-2	-3
4250	-2	-2	-1	-1	-1	-2	-1	-1	-1	-2	-1	-3	-3	-2	-2	-3
4500	-2	-1	0	-1	-1	-1	-1	-1	-1	-2	-1	-3	-3	-3	-2	-4

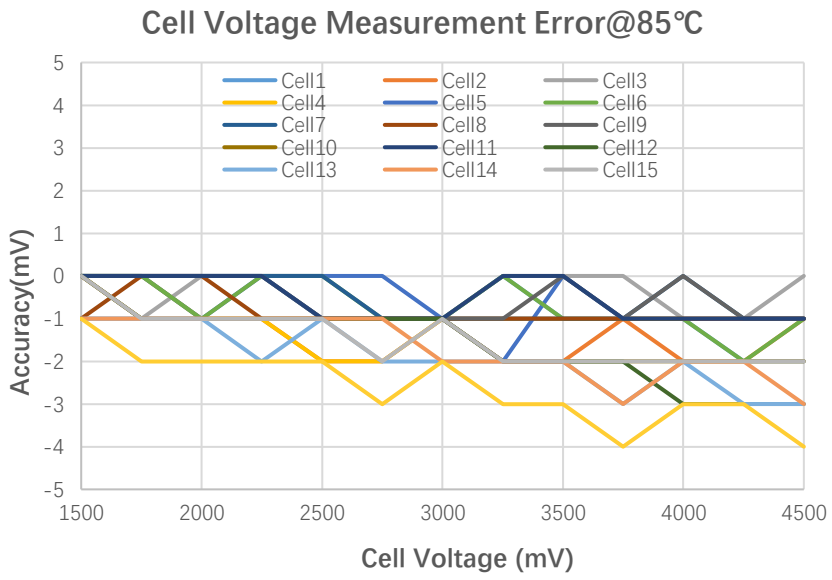


Fig. 5-2-3 电芯电压采样精度@85℃

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5.3 TOS、LD、PACK 电压采样精度

5.3.1 TOS 电压采样精度

Tab.5-3-1 TOS 电压采样精度

VC16 实测电压(V)	TOS(V)	TOS Error (V)
8	8.000	0.000
16	16.000	0.000
32	31.992	-0.008
48	48.000	0.000
59.2	59.200	0.000
64	64.016	0.016
67.2	67.216	0.016
80	80.032	0.032

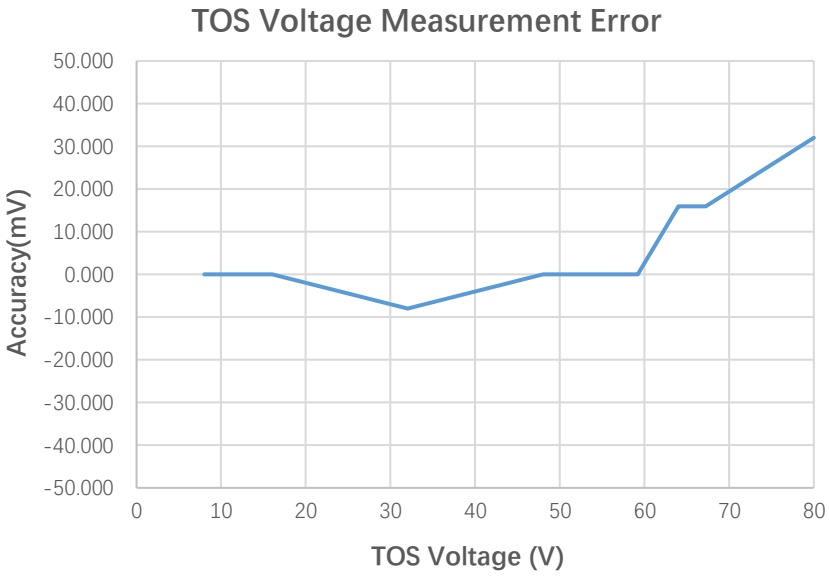


Fig. 5-3-1 TOS电压采样精度

5.3.2 LD 电压采样精度

该系统设计 LD 脚通过 10.2kΩ 电阻连接到 PACK+。

Tab.5-3-2 LD 电压采样精度

LD 实测电压(V)	LD(V)	LD Error (V)
0	0.2	0.2
8	7.976	-0.024
16	15.84	-0.16
32	31.688	-0.312
48	47.52	-0.48
59.2	58.616	-0.584

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64	63.384	-0.616
67.2	66.552	-0.648
80	79.24	-0.76

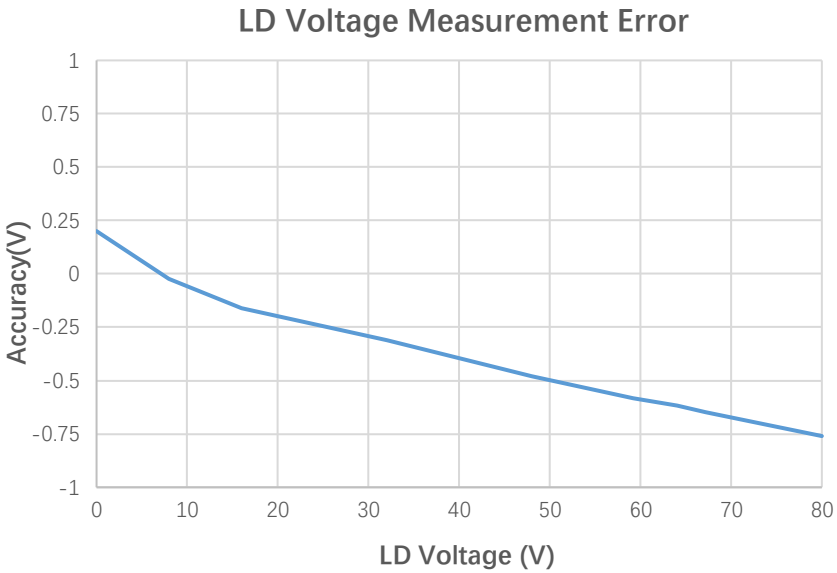


Fig. 5-3-2 LD电压采样精度

5.3.3 PACK 电压采样精度

该系统设计 PACK 脚通过 10.2kΩ 电阻连接到 PACK+。

Tab.5-3-3 PACK 电压采样精度

PACK 实测电压(V)	PACK (V)	PACK Error (V)
0	0.2	0.2
8	7.904	-0.096
16	15.832	-0.168
32	31.672	-0.328
48	47.52	-0.48
59.2	58.616	-0.584
64	63.376	-0.624
67.2	66.536	-0.664
80	79.232	-0.768

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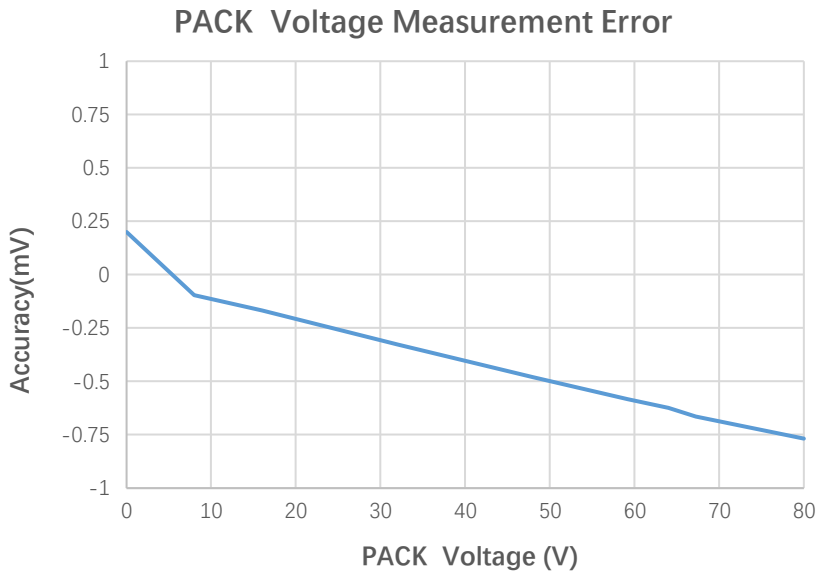


Fig. 5-3-3 PACK电压采样精度

5.4 电池组电流测量精度

Tab.5-4-1 放电电流测量精度@25℃

实测电流(mA)	-25000	-20000	-10000	-5000	-2000	-1000	-500	-200	-100	-50	-20	-10	0
CC2(mA)	-25023	-20009	-10003	-5002	-2002	-999	-499	-199	-99	-50	-20	-10	0
CC2 Error(mA)	-23	-9	-3	-2	-2	1	1	1	1	0	0	0	0

Tab.5-4-2 放电电流测量精度@-40℃

实测电流(mA)	-25000	-20000	-10000	-5000	-2000	-1000	-500	-200	-100	-50	-20	-10	0
CC2(mA)	-25013	-20005	-10002	-5001	-2001	-1000	-500	-201	-101	-51	-20	-10	0
CC2 Error(mA)	-13	-5	-2	-1	-1	0	0	-1	-1	-1	0	0	0

Tab.5-4-3 放电电流测量精度@85℃

实测电流(mA)	-25000	-20000	-10000	-5000	-2000	-1000	-500	-200	-100	-50	-20	-10	0
CC2(mA)	-25030	-20011	-10006	-5003	-2002	-998	-498	-198	-98	-49	-19	-9	1
CC2 Error(mA)	-30	-11	-6	-3	-2	2	2	2	2	1	1	1	1

Tab.5-4-4 充电电流测量精度@25℃

实测电流(mA)	0	10	20	50	100	200	500	1000	2000	5000	10000	20000	25000
CC2(mA)	0	11	21	50	100	200	500	1000	2000	5002	10003	20008	25020
CC2 Error(mA)	0	1	1	0	0	0	0	0	0	2	3	8	20

Tab.5-4-5 充电电流测量精度@-40℃

实测电流(mA)	0	10	20	50	100	200	500	1000	2000	5000	10000	20000	25000
CC2(mA)	0	10	20	50	100	199	500	1000	2000	5001	10002	20005	25010
CC2 Error(mA)	0	0	0	0	0	-1	0	0	0	1	2	5	10

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Tab.5-4-6 充电电流测量精度@85℃

实测电流(mA)	0	10	20	50	100	200	500	1000	2000	5000	10000	20000	25000
CC2(mA)	1	12	21	49	100	200	499	998	2002	5003	10006	20013	25033
CC2 Error(mA)	1	2	1	-1	0	0	-1	-2	2	3	6	13	33

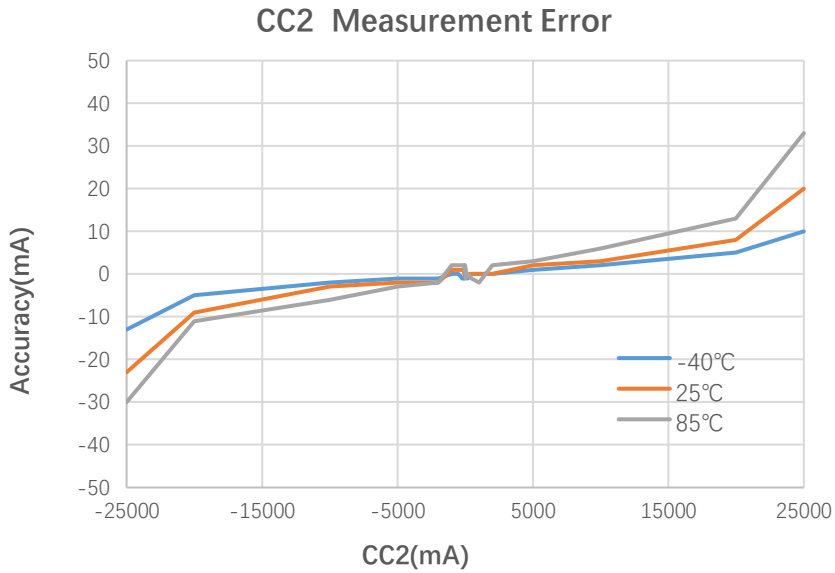


Fig. 5-4-1 充放电电流测量精度

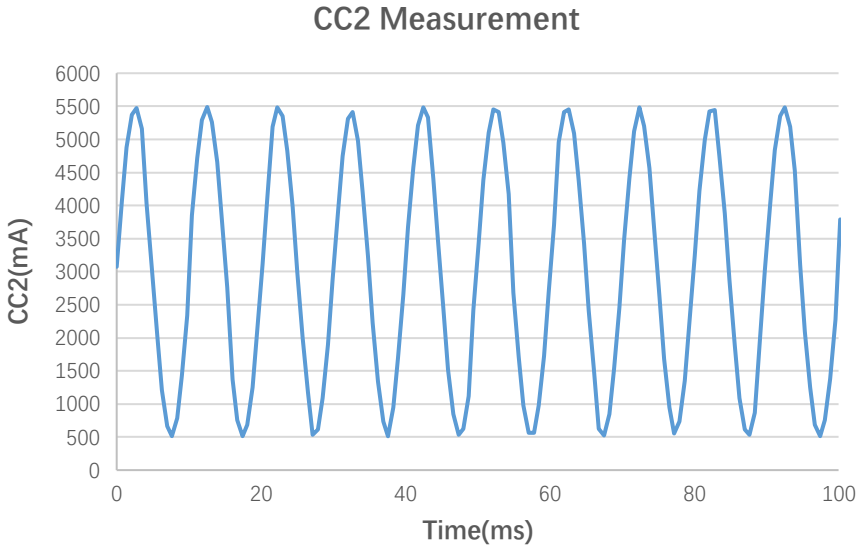


Fig. 5-4-2 100Hz 正弦波放电电流(3A DC+5A AC)测量

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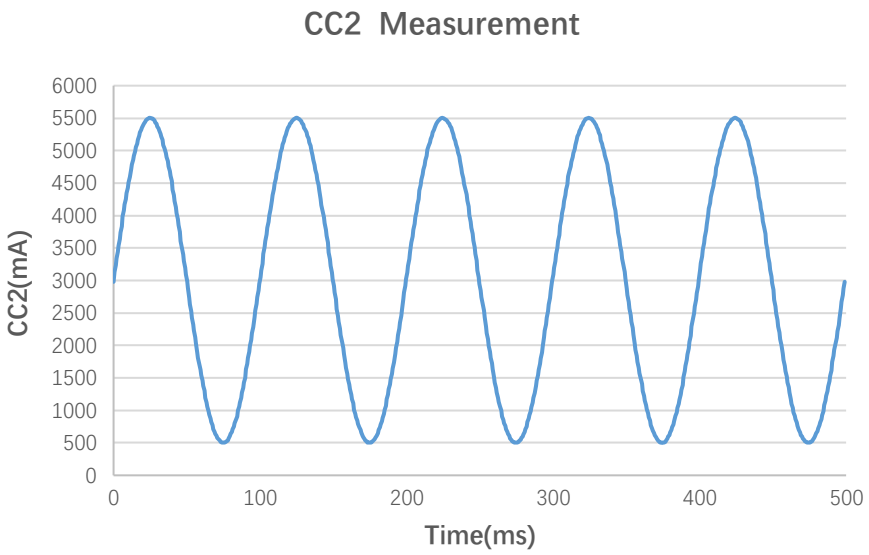


Fig. 5-4-3 10Hz正弦波放电电流(3A DC+5A AC)测量

5.5 温度测量精度

本系统设计采用 AT103-2 热敏电阻，温度采集数据经过 MCU 查表法校准。具体操作方法请参阅 GD30BM2016 温度测量优化。

Tab.5-5-1 温度采样精度

实测温度(℃)	-40	-30	-20	-10	0	10	20	25	30	40	50	60	70	80	85	90	100	110	120
TS(℃)	-39.7	-30.5	-19.9	-9.6	-0.2	9.7	19.3	24.4	29.8	40.7	51.2	60.6	70.4	79.3	84.6	90.1	100.8	110.6	119.2
TS Error(℃)	0.3	-0.5	0.1	0.4	-0.2	-0.3	-0.7	-0.6	-0.2	0.7	1.2	0.6	0.4	-0.7	-0.4	0.1	0.8	0.6	-0.8

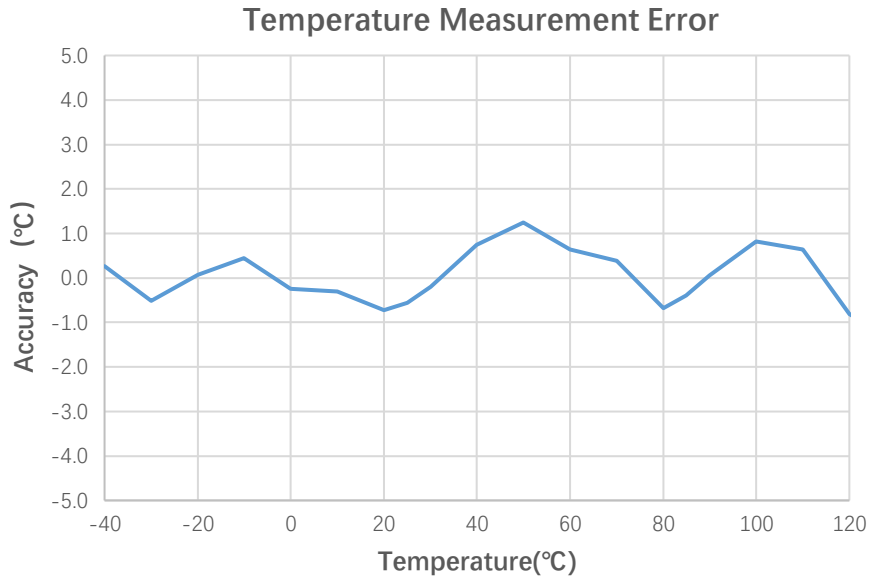


Fig. 5-5-1 温度测量精度

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5.6 REG 输出

本系统设计器件 GD30BM2016 具有四路 LDO，包括 REG18、Pre-REG、REG1、REG2。

5.6.1 REG18 输出

Tab.5-6-1 REG18 线性调整率和负载调整率

VBAT(V)	VREG18(V)							负载调整率(mV/mA)
	Io=0mA	Io=1mA	Io=2mA	Io=3mA	Io=4mA	Io=5mA	Io=6mA	
8	1.802	1.801	1.800	1.800	1.799	1.798	1.798	0.67
59.2	1.801	1.801	1.800	1.799	1.799	1.798	1.797	0.67
80	1.801	1.800	1.800	1.799	1.798	1.798	1.797	0.67
线性调整率(mV/V)	0.014	0.014	0	0.014	0.014	0	0.014	/

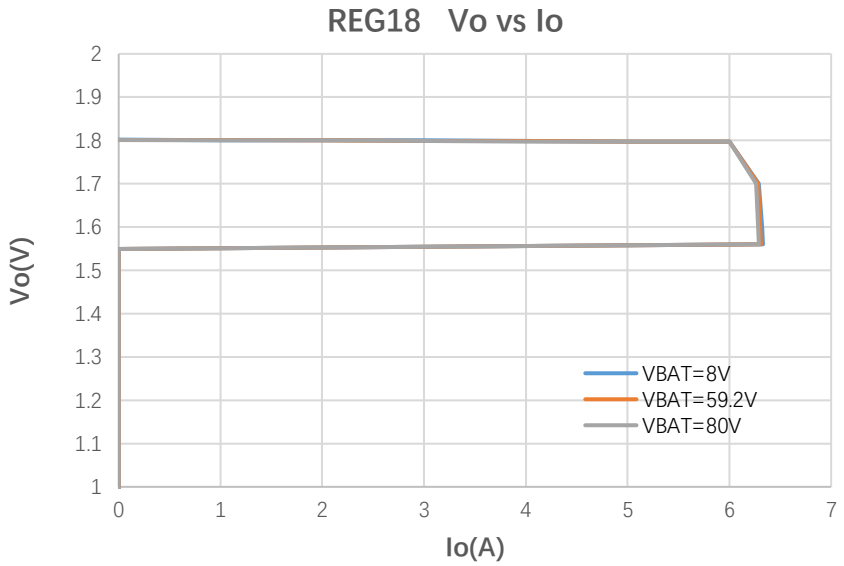
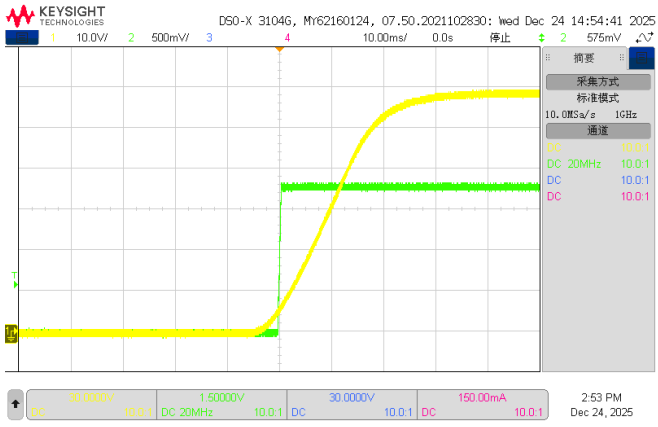
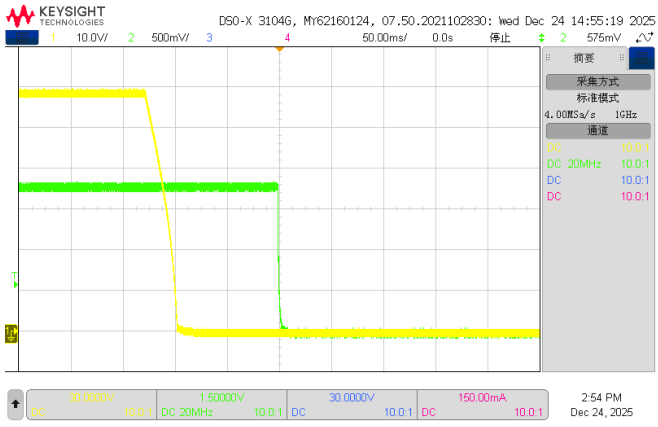


Fig.5-6-1 REG18 输出电压和电流曲线



CH1:BAT CH2:REG18

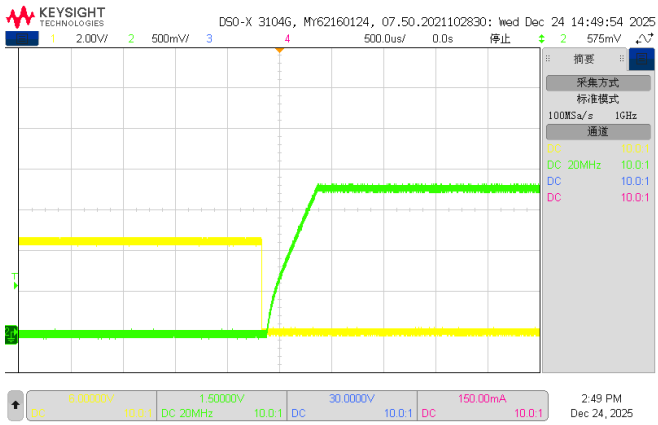


CH1:BAT CH2:REG18

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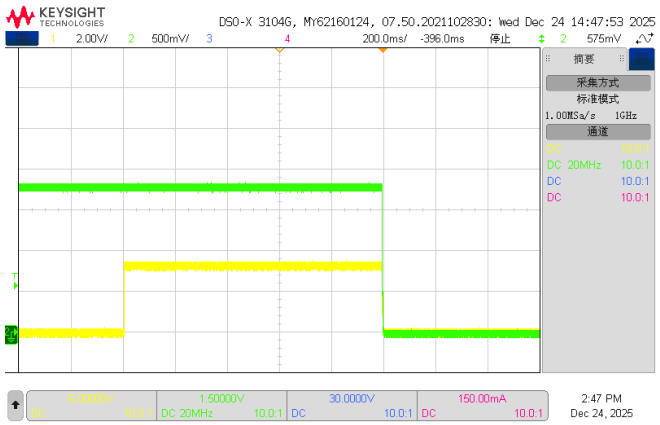
Fig.5-6-2 VBAT 上电



CH1:TS2 CH2:REG18

Fig.5-6-4 WAKE UP 启动

Fig.5-6-3 VBAT 下电



CH1:RST_SHUT CH2:REG18

Fig.5-6-5 RST_SHUT 关闭

5.6.2 Pre-REG 输出

Tab.5-6-2 Pre-REG 输出电压和短路电流

VBAT(V)	Pre-REG (V)						ISC(mA)
	Io=0mA	Io=20mA	Io=40mA	Io=60mA	Io=80mA	Io=100mA	
16	5.524	5.518	/	/	/	/	39.3
32	5.524	5.518	5.513	5.363	/	/	75.3
59.2	5.525	5.519	5.514	5.508	5.497	5.456	136.5
80	5.524	5.518	5.513	5.507	5.502	5.495	182.5

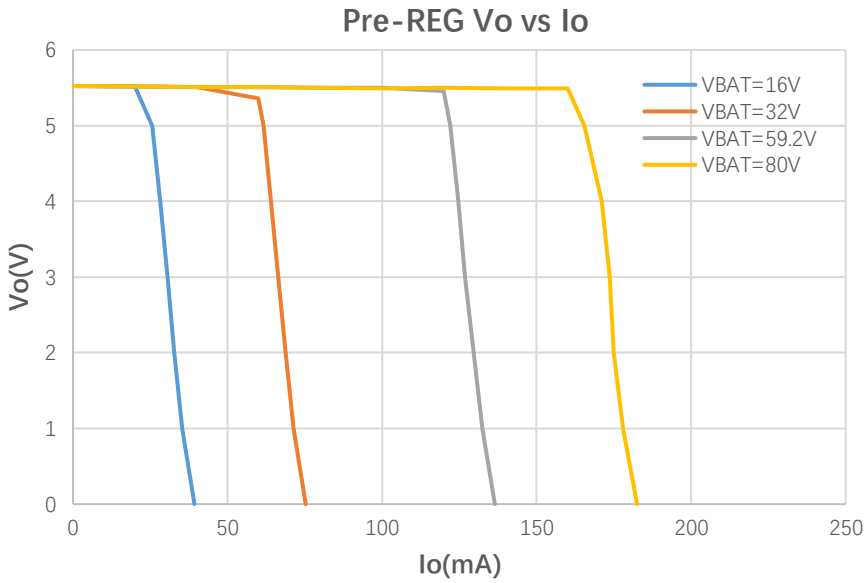


Fig.5-6-6 Pre-REG 输出电压和电流曲线

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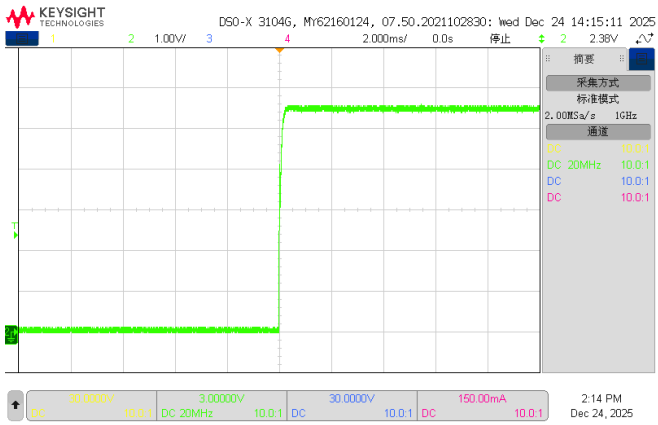


Fig.5-6-7 Pre-REG 使能@BAT=59.2V

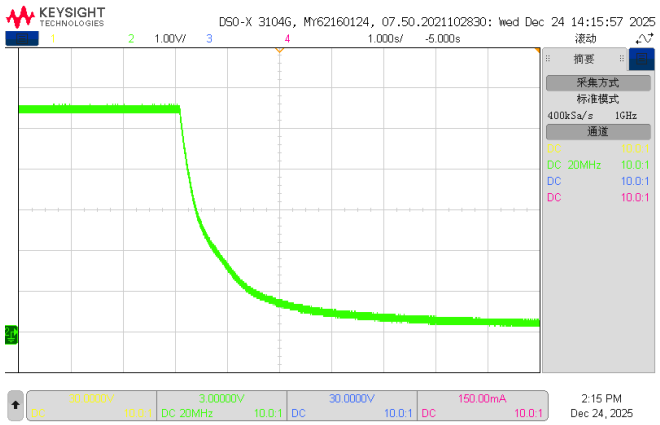


Fig.5-6-8 Pre-REG 禁能@BAT=59.2V

5.6.3 REG1 输出

Tab.5-6-3 REG1 输出电压和短路电流

REG1 输出配置	VREG1(V)						ISC(mA)
	Io=0mA	Io=10mA	Io=20mA	Io=30mA	Io=40mA	Io=45mA	
REG1_1.8	1.791	1.785	1.78	1.775	1.769	1.767	58.5
REG1_2.5	2.493	2.486	2.481	2.476	2.470	2.467	58.5
REG1_3.0	3.015	3.007	3.002	2.996	2.991	2.988	58.5
REG1_3.3	3.291	3.283	3.277	3.272	3.266	3.263	58.5
REG1_5.0	5.057	5.048	5.042	5.036	5.028	/	58.5

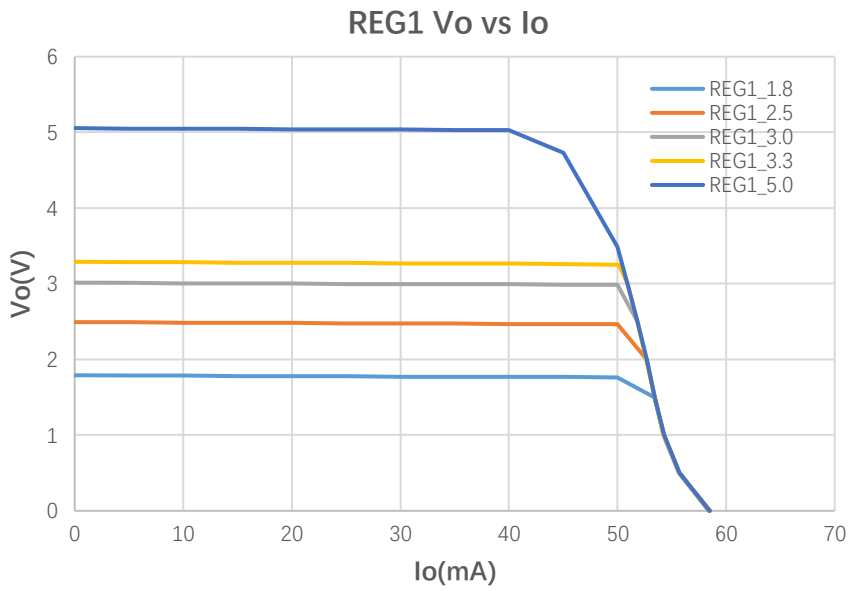


Fig.5-6-9 REG1 输出电压和电流曲线

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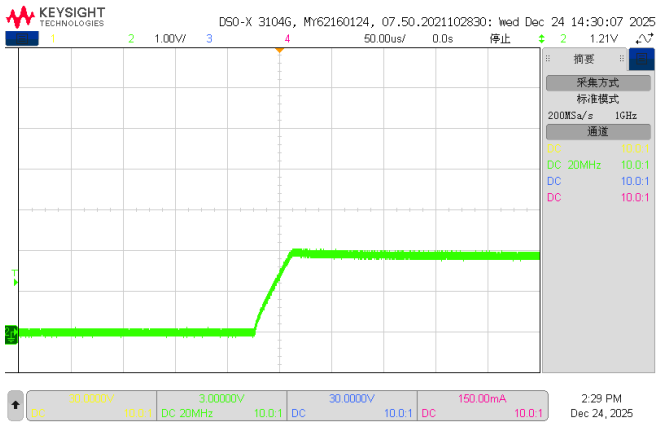


Fig.5-6-10 REG1_1.8 使能

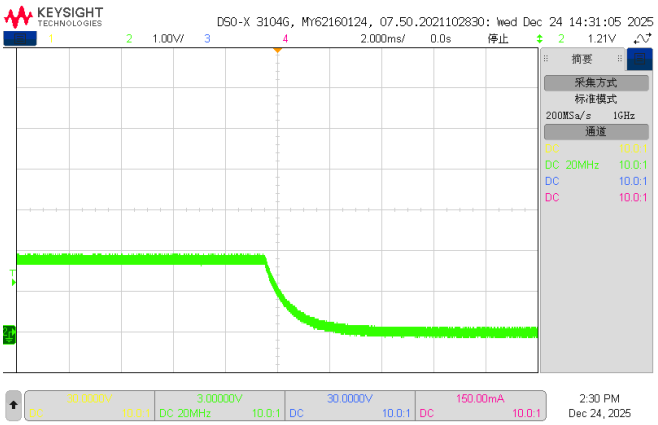


Fig.5-6-11 REG1_1.8 禁能

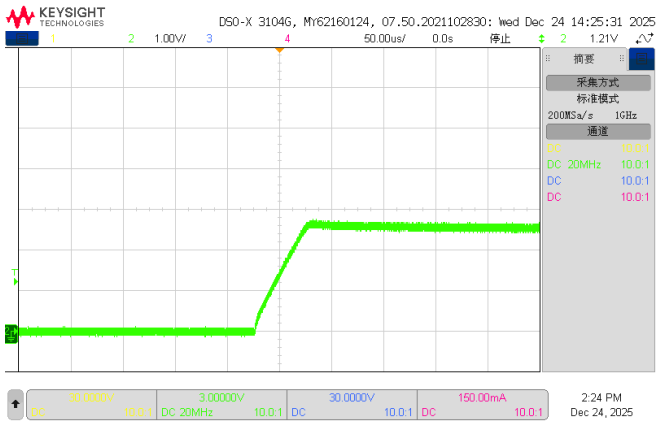


Fig.5-6-12 REG1_2.5 使能

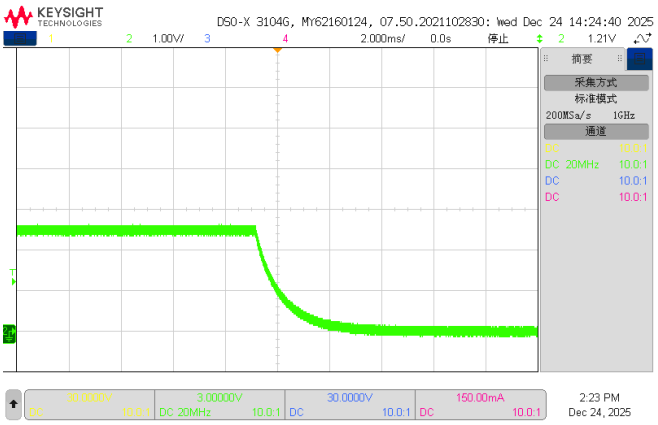


Fig.5-6-13 REG1_2.5 禁能

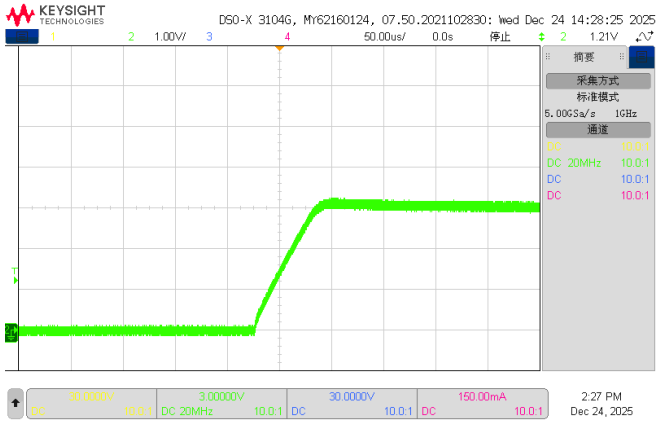


Fig.5-6-13 REG1_3.0 使能

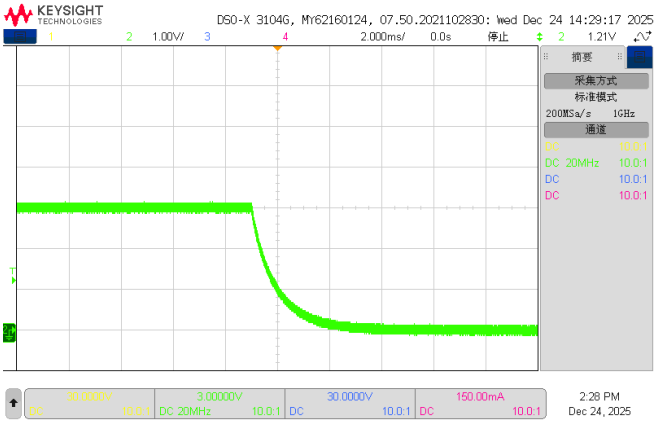


Fig.5-6-14 REG1_3.0 禁能

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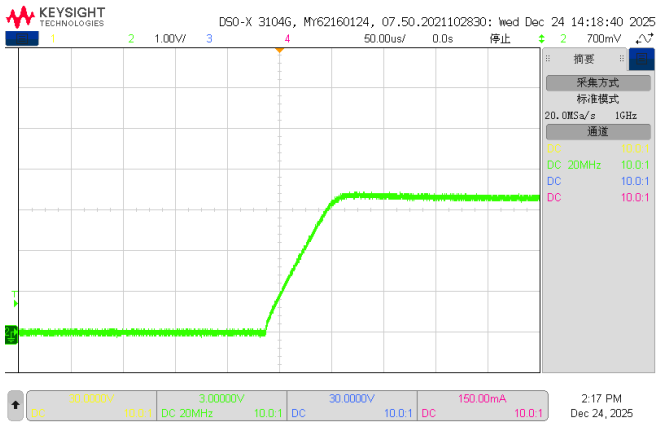


Fig.5-6-15 REG1_3.3 使能

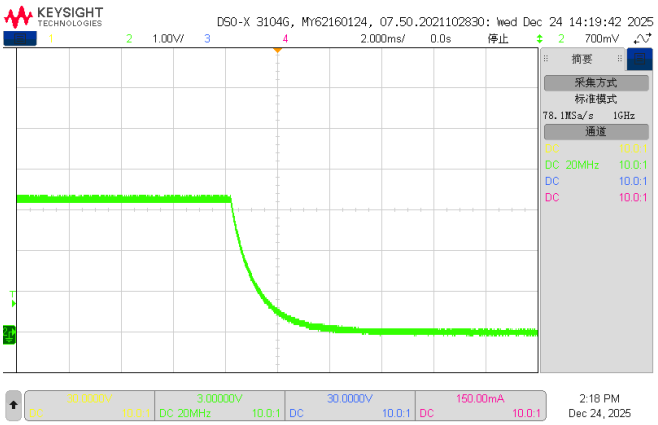


Fig.5-6-16 REG1_3.3 禁能

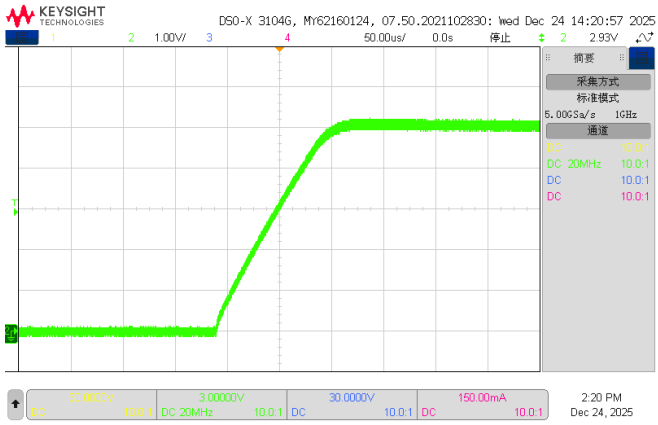


Fig.5-6-17 REG1_5.0 使能

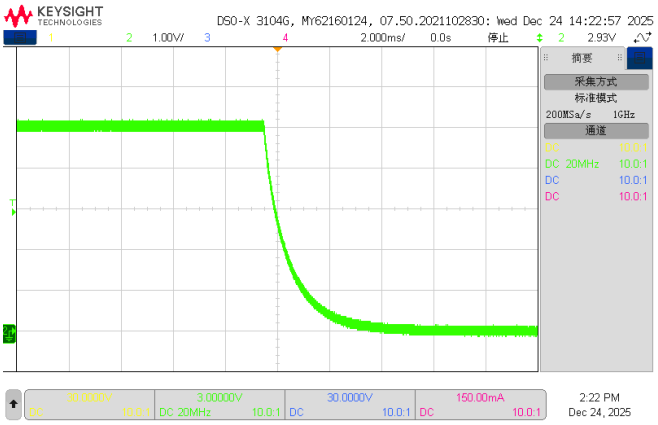


Fig.5-6-18 REG1_5.0 禁能

5.6.4 REG2 输出

Tab.5-6-4 REG2 输出电压和短路电流

VREG1 输出配置	VREG2(V)						ISC(mA)
	Io=0mA	Io=10mA	Io=20mA	Io=30mA	Io=40mA	Io=45mA	
VREG2_1.8	1.821	1.815	1.811	1.806	1.801	1.799	57.1
VREG2_2.5	2.548	2.541	2.536	2.532	2.527	2.524	57.1
VREG2_3.0	3.094	3.087	3.082	3.077	3.071	3.069	57.1
VREG2_3.3	3.384	3.377	3.372	3.367	3.362	3.359	57.1
VREG2_5.0	5.275	5.267	5.260	5.254	5.007	/	57.1

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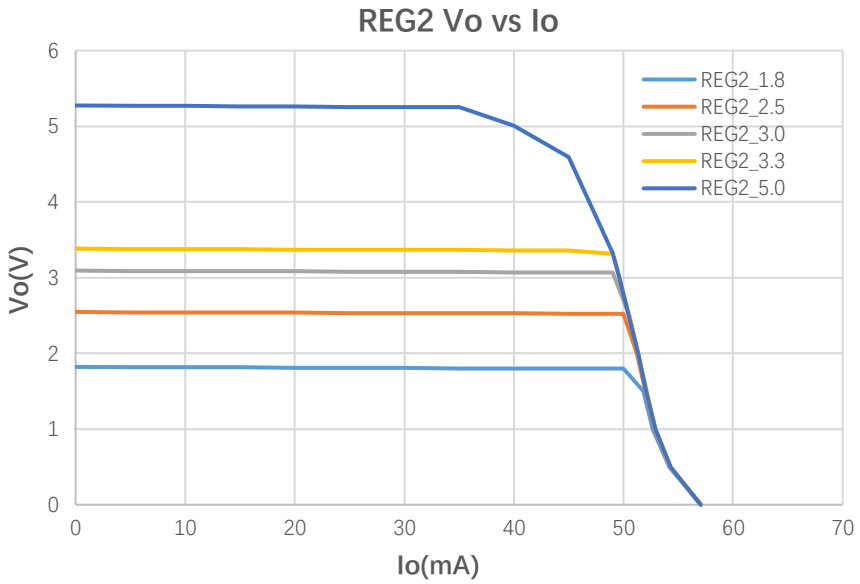


Fig.5-6-19 REG2 输出电压和电流曲线

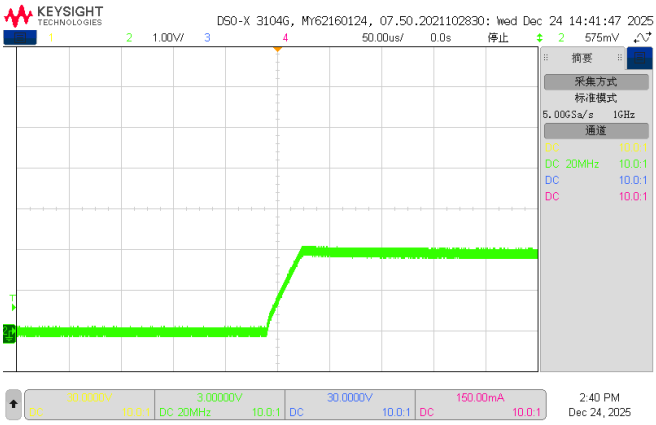


Fig.5-6-20 REG2_1.8 使能

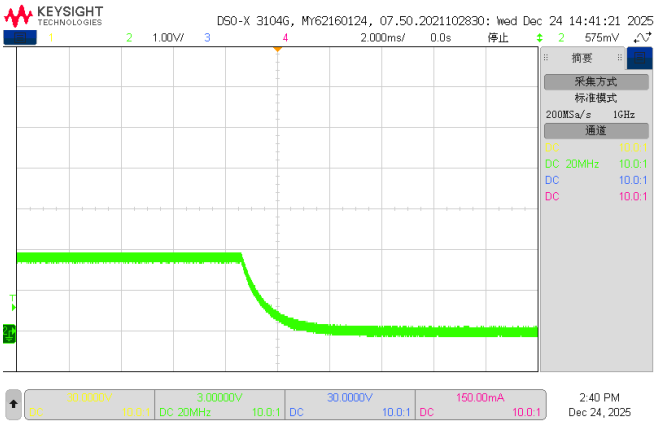


Fig.5-6-21 REG2_1.8 禁能

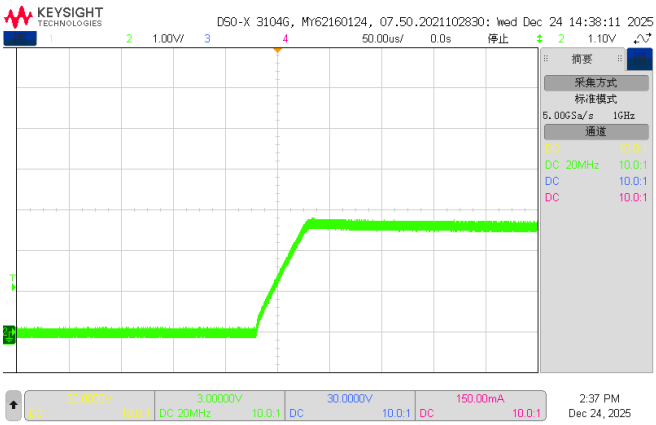


Fig.5-6-22 REG2_2.5 使能

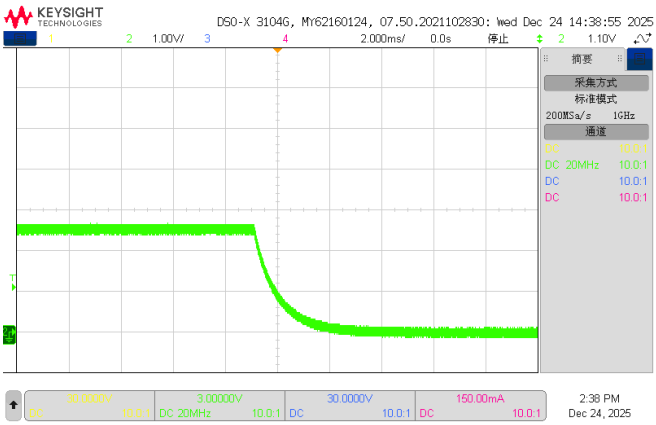


Fig.5-6-23 REG2_2.5 禁能

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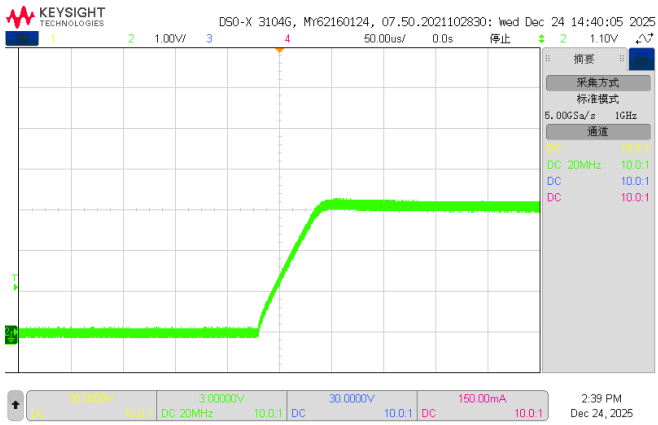


Fig.5-6-24 REG2_3.0 使能

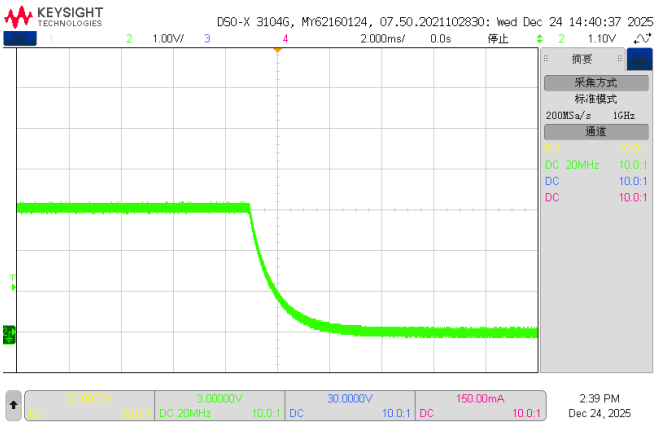


Fig.5-6-25 REG2_3.0 禁能

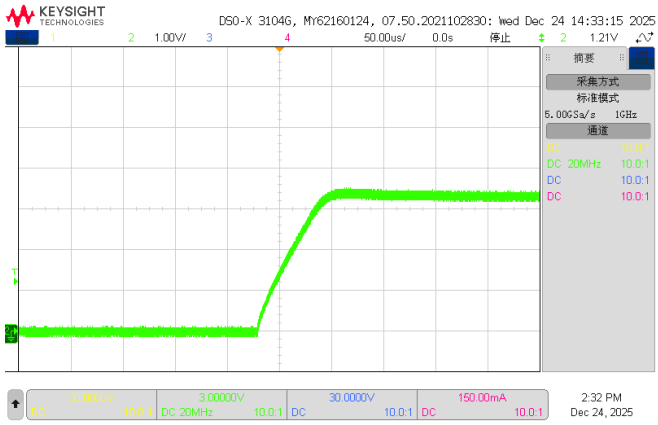


Fig.5-6-26 REG2_3.3 使能

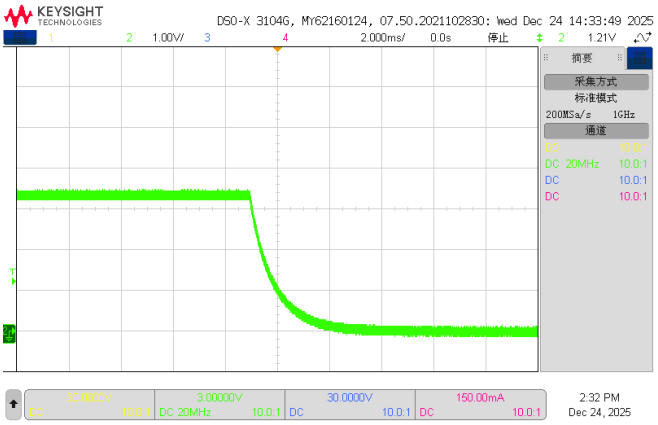


Fig.5-6-27 REG2_3.3 禁能

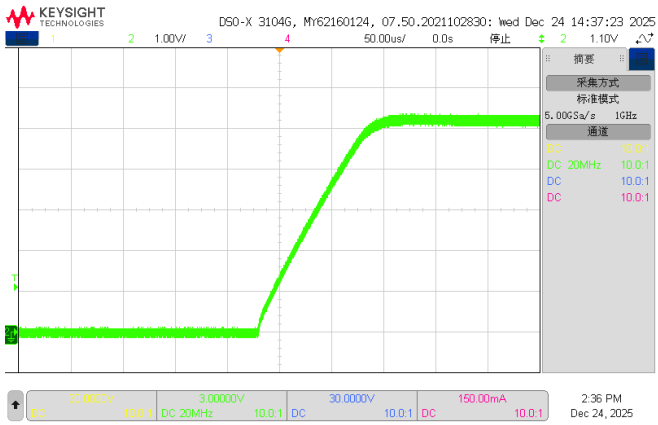


Fig.5-6-28 REG2_5.0 使能

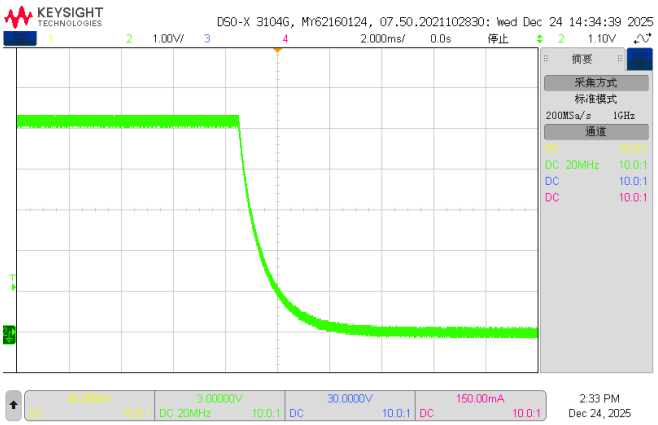


Fig.5-6-29 REG2_5.0 禁能

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5.7 CP 输出

本系统设计器件 GD30BM2016 具有电荷泵功能，可配置 5V 和 11V 输出，用于高边 NMOS 驱动。

Tab.5-7-1 CP 输出电压带载能力@VBAT=57.6V

CP 输出配置	VCP(V)					Isc(uA)
	Io=0uA	Io=20uA	Io=40uA	Io=60uA	Io=100uA	
CP_5V	4.9	4.9	4.9	4.9	5.0	53
CP_11V	10.6	10.6	9.52	8.5	6.1	53

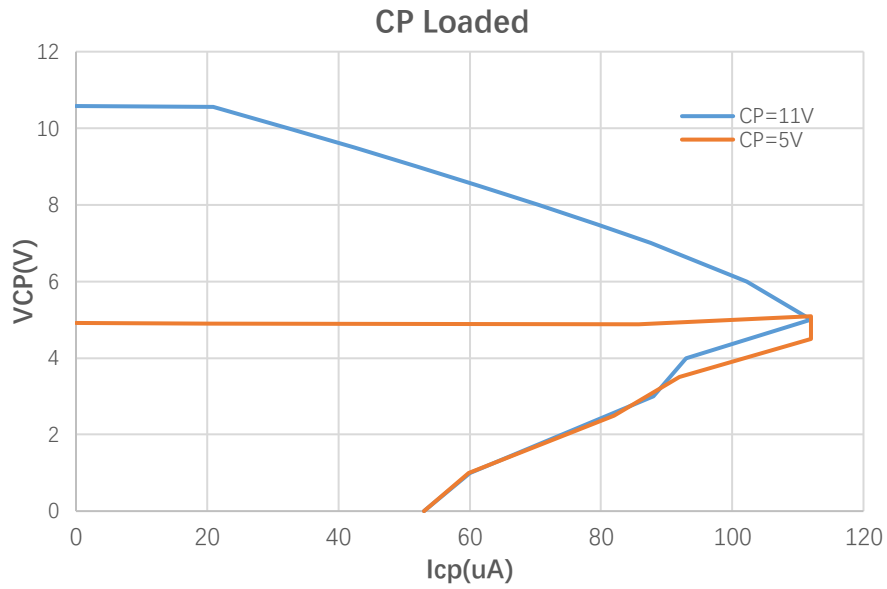
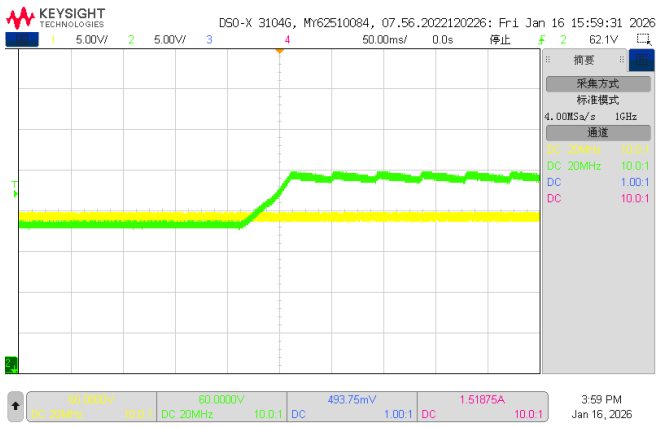
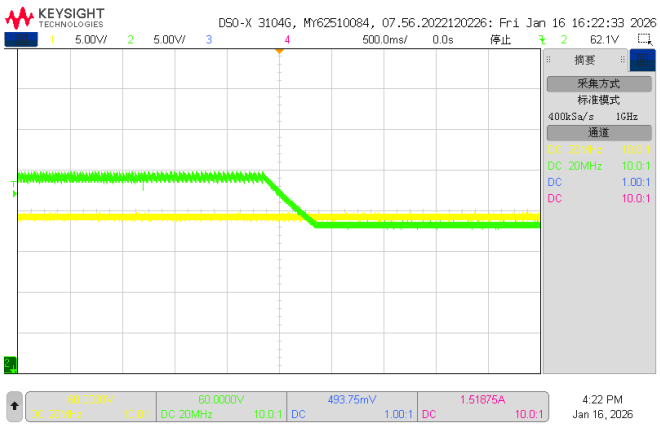


Fig.5-7-1 CP 带载能力@VBAT=57.6V



CH1: BAT CH2:CP1

Fig.5-7-2 CP_5.0V 使能

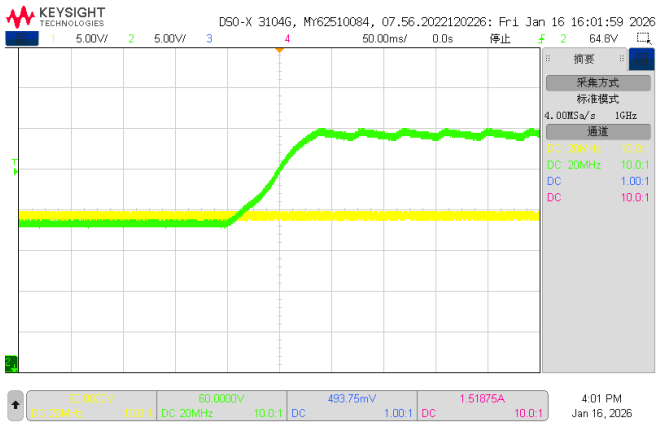


CH1: BAT CH2:CP1

Fig.5-7-3 CP_5.0V 禁能

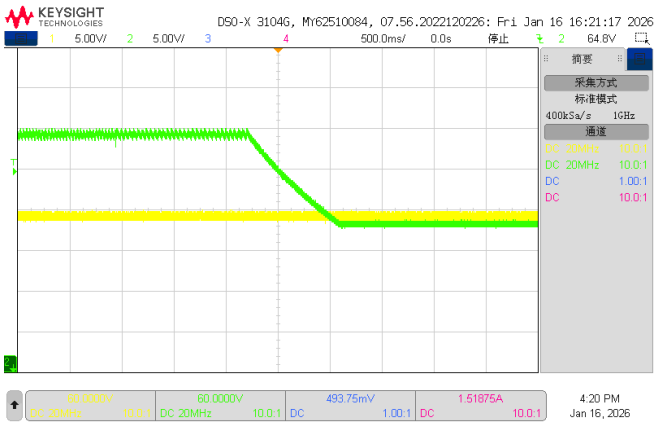
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CH1: BAT+ CH2:CP1

Fig.5-7-4 CP_11V 使能



CH1: BAT+ CH2:CP1

Fig.5-7-5 CP_11V 禁能

5.8 驱动

5.8.1 DSG 驱动

Tab.5-8-1 DSG 驱动带载能力

IDSG(uA)	DSG (V)	CHG (V)
0	46.05	46.1
20	45.97	45.98
40	45.42	45.39
60	44.37	44.34
80	43.24	43.18
100	41.88	41.82
90.7	40	40.06
38.6	36	36.61
773	32	33.8
2406	30	30.1
7336	25	31.65
11970	20	30.56
17550	10	29.59
20460	0	29.59

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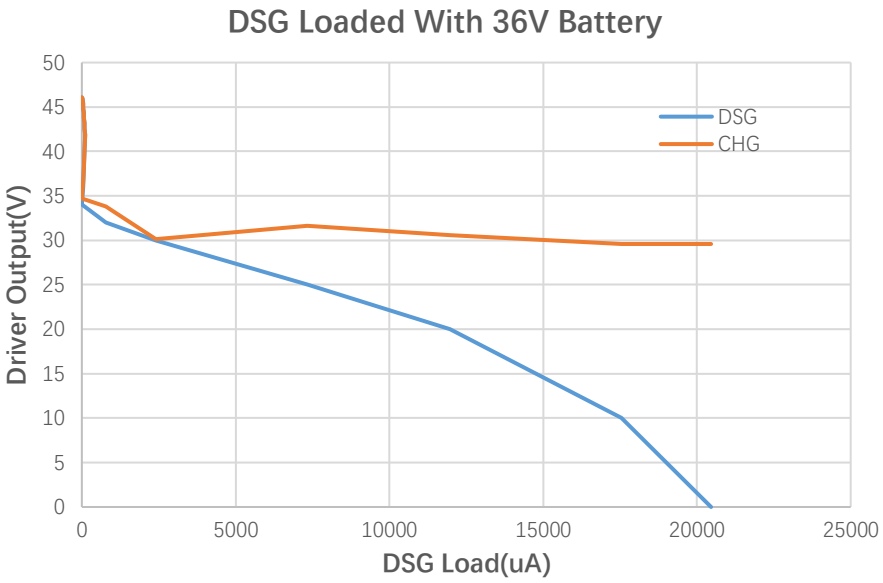
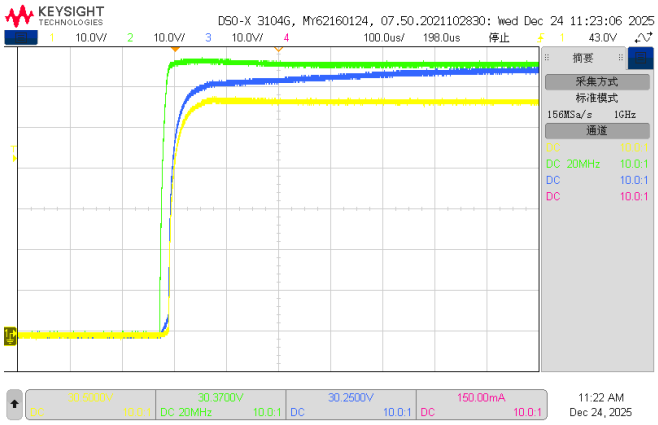
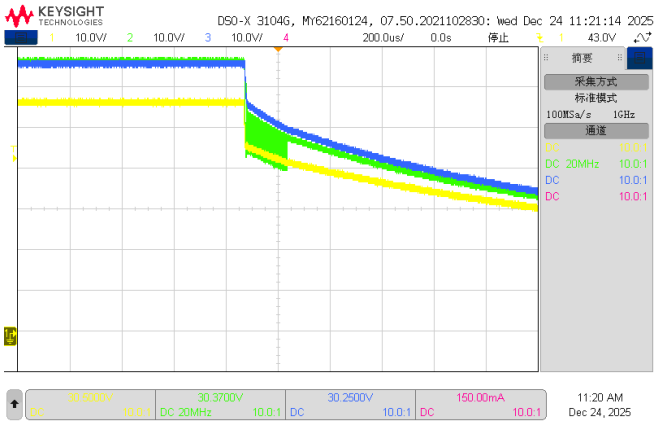


Fig.5-8-1 DSG 带载能力



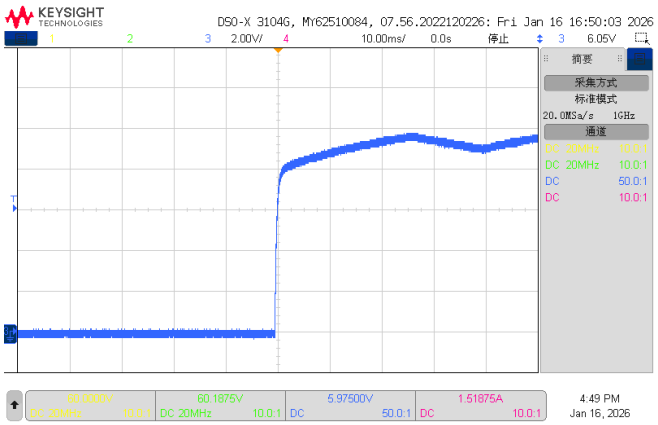
CH1:LD CH2:DSG CH3:DSG_G

Fig.5-8-2 DSG On @BAT=59.2



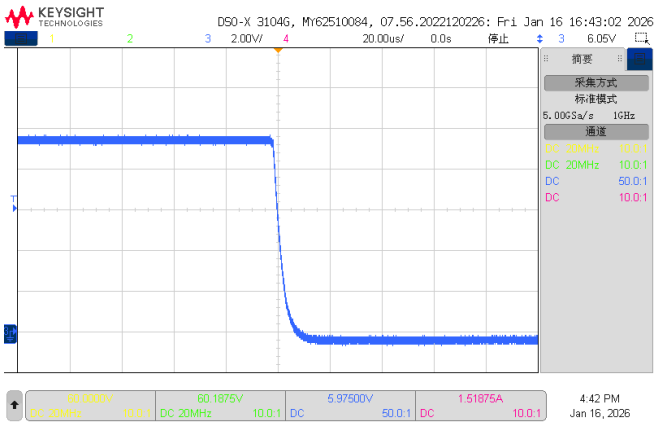
CH1:LD CH2:DSG CH3:DSG_G

Fig.5-8-3 DSG Off @BAT=59.2



CH3:DFET_GS

Fig.5-8-4 DSG On @BAT=59.2



CH3:DFET_GS

Fig.5-8-5 DSG Off @BAT=59.2

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5.8.2 CHG 驱动

Tab.5-8-2 CHG 驱动带载能力

ICHG(uA)	CHG (V)	DSG (V)
0	46.10	46.05
20	45.96	45.98
40	45.40	45.42
60	44.36	44.35
80	43.23	43.2
100	41.84	41.87
90.3	40	40.04
26.3	36	35.33
3944	32	32.99
7511	30	32.07
15510	25	30.09
20330	20	28.99
20500	10	28.94
19530	0	29.12

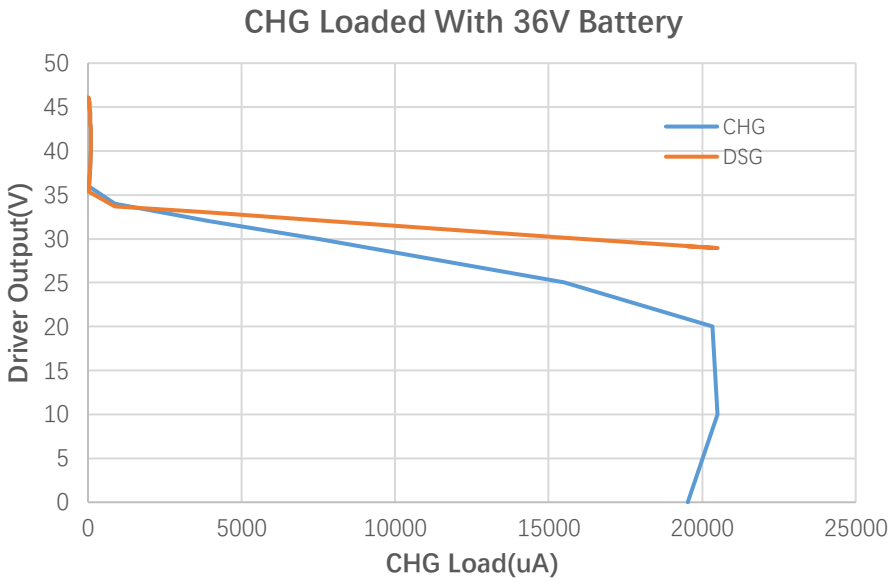
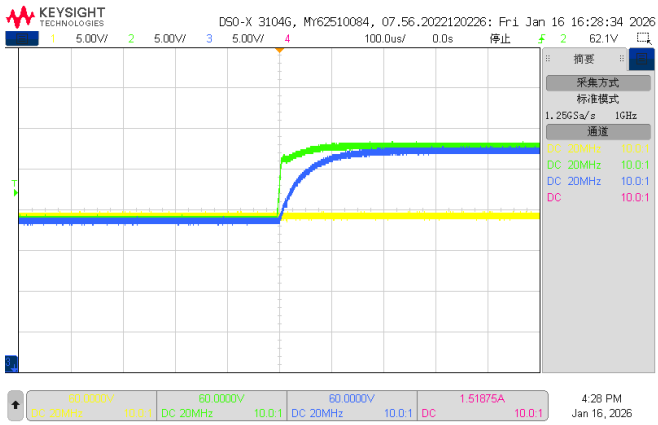


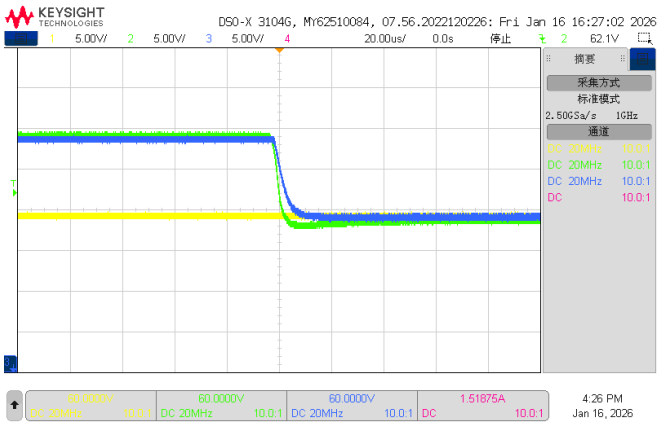
Fig.5-8-6 CHG 带载能力

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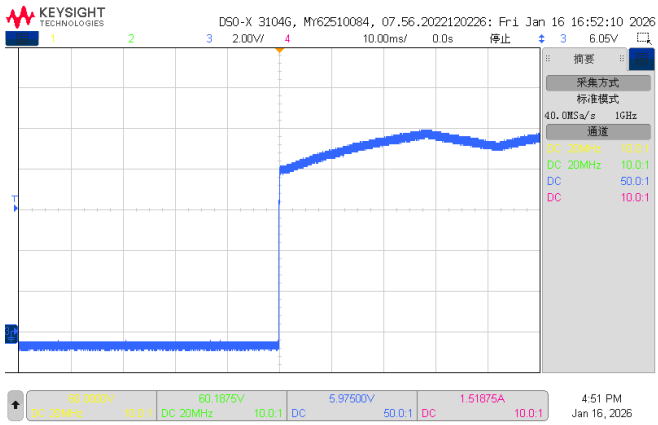
CH1:BAT CH2:CHG CH3:CHG-G

Fig.5-8-7 CHG On @BAT=59.2V



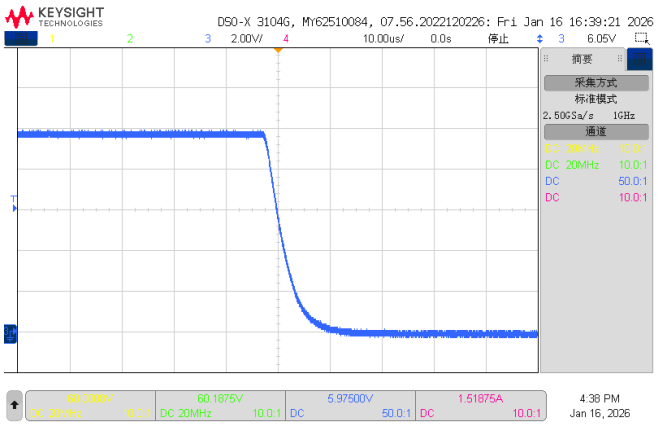
CH1:BAT CH2:CHG CH3:CHG-G

Fig.5-8-8 CHG Off @BAT=59.2V



CH3:CFET_GS

Fig.5-8-9 CHG On @BAT=59.2V



CH3:CFET_GS

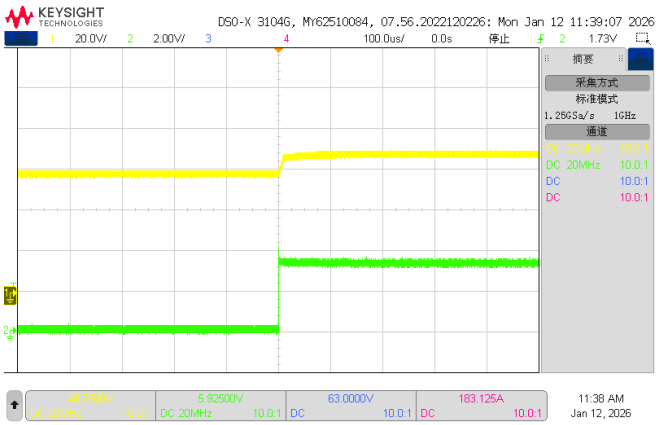
Fig.5-8-10 CHG Off @BAT=59.2V

5.8.3 高低边驱动同步

DCHG/DDSG 引脚配置成 DCHG/DDSG 功能，用于映射 CHG/DSG 状态。可用于 MCU 检测 AFE 故障中断，及低边驱动开关。

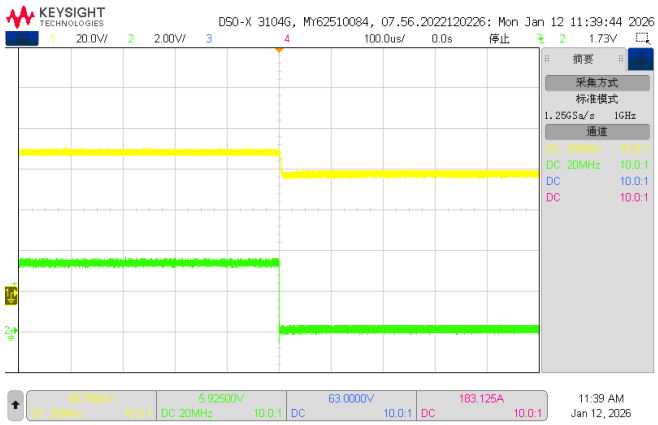
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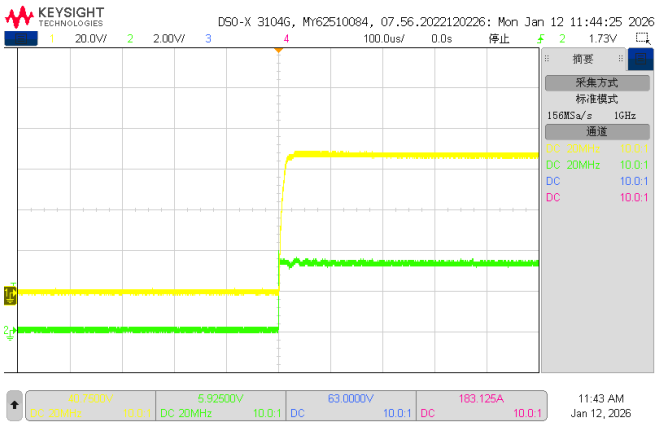
CH1:CHG CH2:DCHG

Fig.5-8-11 CHG/DCHG On @BAT=59.2V



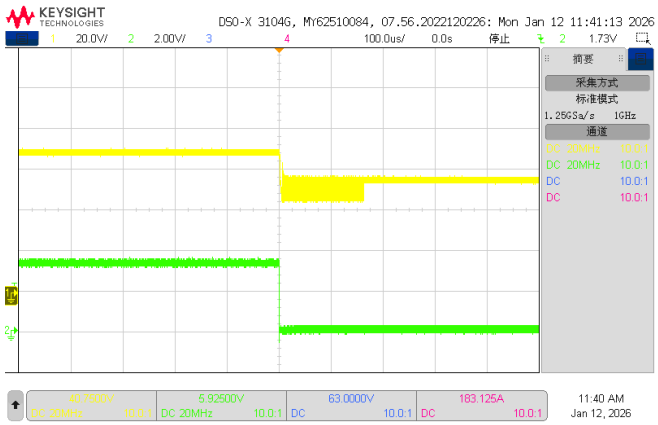
CH1:CHG CH2:DCHG

Fig.5-8-12 CHG/DCHG OFF @BAT=59.2V



CH1:DSG CH2:DDSG

Fig.5-8-13 DSG/DDSG On @BAT=59.2V

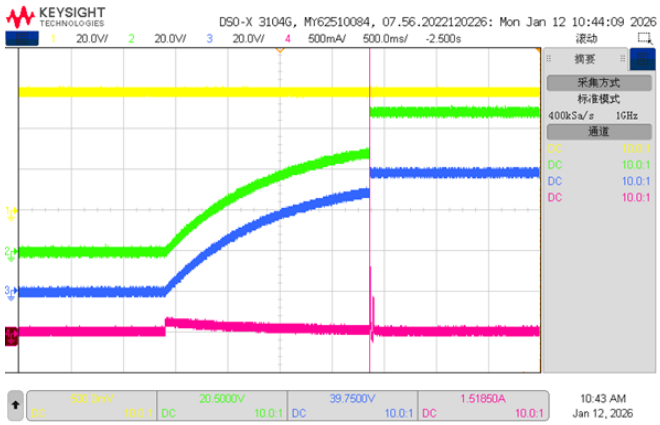


CH1:DSG CH2:DDSG

Fig.5-8-14 DSG/DDSG OFF @BAT=59.2V

5.9 预放电转换

本系统设计有预防电功能，当 DSG FET 开启前，先开启 PDSG PFET，从而减少负载最初上电时的浪涌电流。



CH1:BAT CH2:DSG CH3:PACK CH4:ILOAD

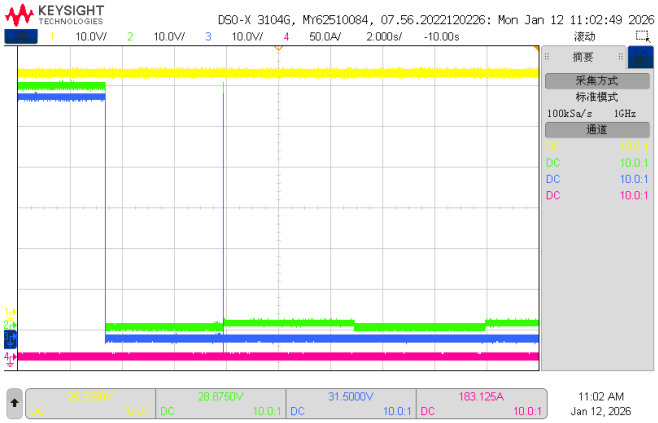
Fig.5-9-1 Predischarge @BAT=59.2V，CL=2200uF

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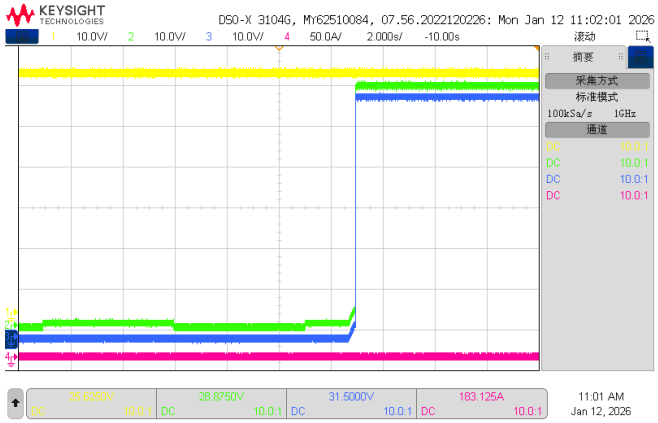
5.10 负载检测功能

本系统设计配置有负载检测功能，当 GD30BM2016 发生 OCDL/SCDL 时进入，当输出阻抗超过 30kΩ 时退出(LD 电压超过 4V)。



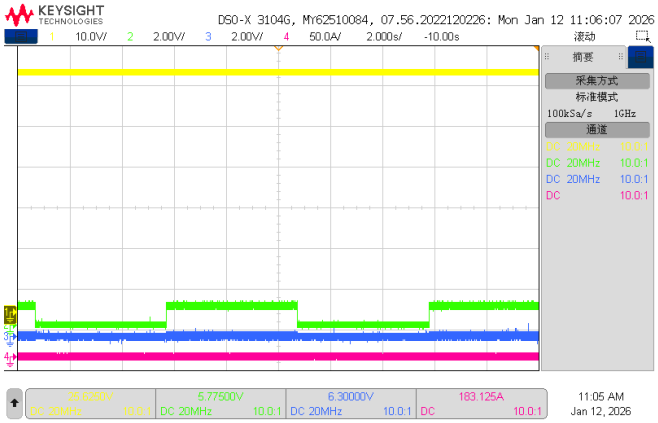
CH1:BAT CH2:LD CH3:PACK CH4:ILOAD

Fig.5-10-1 放电短路进入负载检测功能 @BAT=59.2V



CH1:BAT CH2:LD CH3:PACK CH4:ILOAD

Fig.5-10-2 放电短路移除恢复输出 @BAT=59.2V



CH1:BAT CH2:LD CH3:PACK CH4:ILOAD

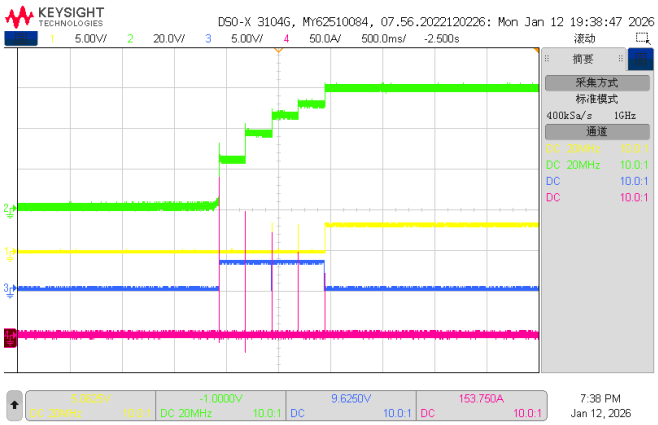
Fig.5-10-3 负载检测功能 @BAT=59.2V

5.11 容性负载启动

容性负载启动时，容易触发 SCD，需要重启多次给电容充电，由于 GD30BM2016 SCD 重启间隔步长 1s，所以 PACK+建立时间较长。为了解决这个问题，可以将 SCD 间隔配置成 0，利用 MCU 检测 DDSG 信号，当 DDSG 变低时，MCU 给 DFETOFF 250ms 的高电平（DFETOFF 配置高电平有效），用于关闭 DSG，250ms 后如无其他故障，DSG 恢复开启。这样相当于 SCD 重启间隔设置到 250ms，PACK 建立时间缩短。

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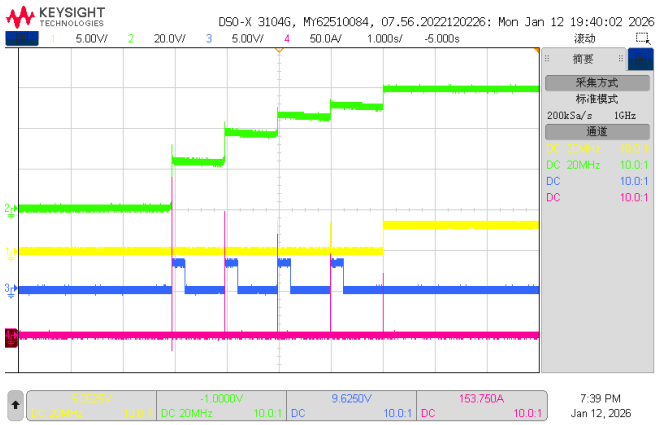
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CH1:DDSG CH2:PACK+ CH3:DFETOFF CH4:ILOAD

Fig.5-11-1 容性负载启动 @BAT=59.2V，CL=2200uF

MCU 实现 SCD 重启间隔 250ms



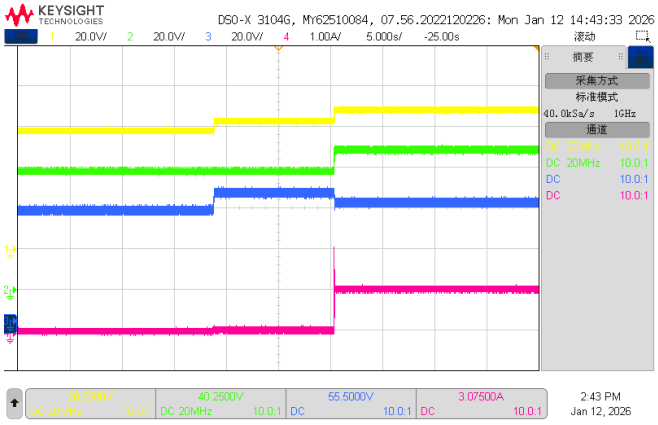
CH1:DDSG CH2:PACK+ CH3:DFETOFF CH4:ILOAD

Fig.5-11-1 容性负载启动 @BAT=59.2V，CL=2200uF

AFE 配置 SCD 重启间隔 1000ms

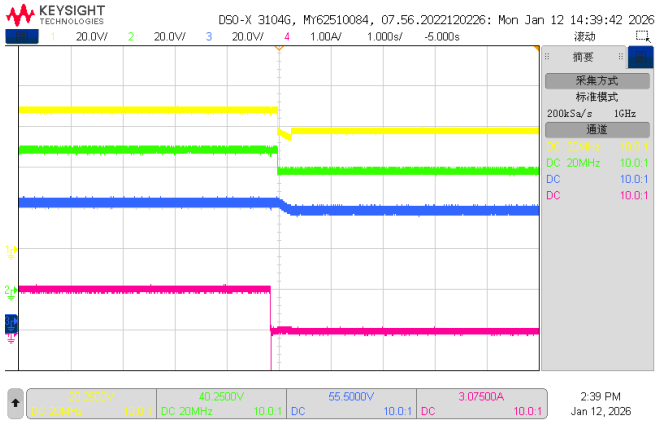
5.12 工作模式切换

当电池组充电或放电时，它处于正常模式。当不充电或不放电时，电池组处于睡眠模式，正在等待充电器或负载连接。睡眠模式唤醒时报Alert,可用于唤醒MCU。



CH1:DSG CH2:CHG CH3:PACK CH4:ICharge

Fig.5-12-1 睡眠模式切换到正常模式（充电）

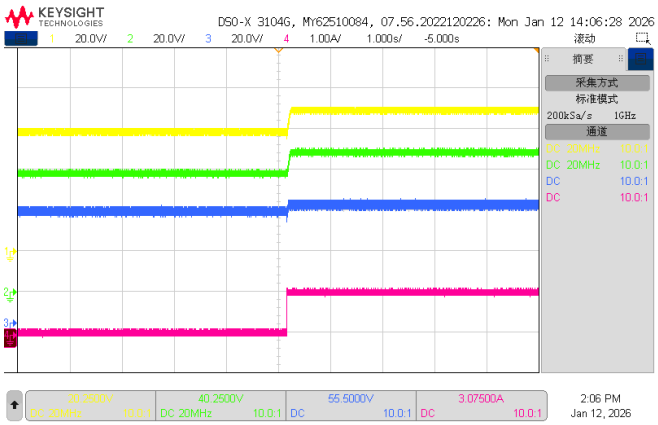


CH1:DSG CH2:CHG CH3:PACK CH4: ICharge

Fig.5-12-2 正常模式(充电)切换到睡眠模式

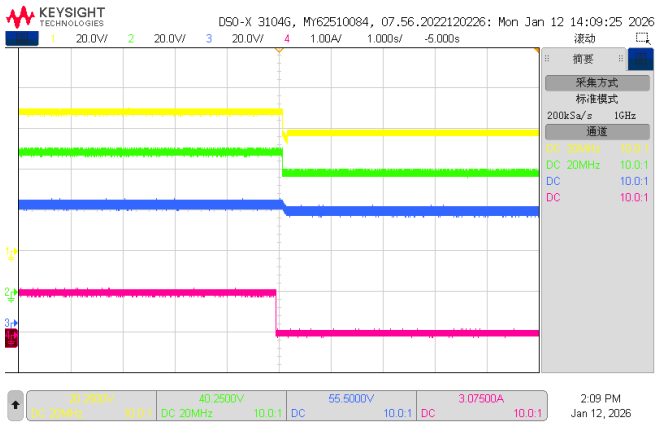
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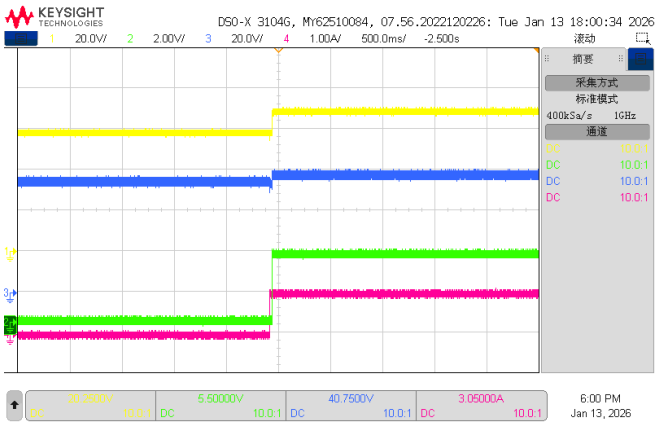
CH1:DSG CH2:CHG CH3:PACK CH4:ILOAD

Fig.5-12-3 睡眠模式切换到正常模式（放电）



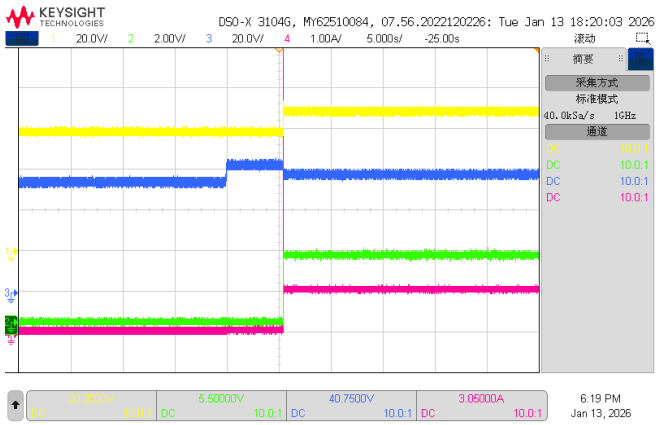
CH1:DSG CH2:CHG CH3:PACK CH4:ILOAD

Fig.5-12-4 正常模式(放电)切换到睡眠模式



CH1:DSG CH2:Alert CH3:PACK CH4:ILOAD

Fig.5-12-5 睡眠模式唤醒 Alert 告警（放电）



CH1:CHG CH2:Alert CH3:PACK CH4: ICharge

Fig.5-12-6 睡眠模式唤醒 Alert 告警（充电）

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5.13 电芯均衡

本系统设计采用了外部电芯均衡。在 3.6V 电芯电压下，峰值均衡电流为 83mA，大概 68mA 流经外部均衡电阻器，并有约 15mA 流经 GD30BM2016 器件。

Tab.5-13-1 电芯均衡占空比及均衡电流

Loop_Slow	均衡占空比(%)	均衡电流(mA)	均衡时间(ms)
Full speed	/	/	/
Half speed	51.4	83	45.6
1/4 speed	75.7	83	136.7
1/8 speed	88.1	83	318.9

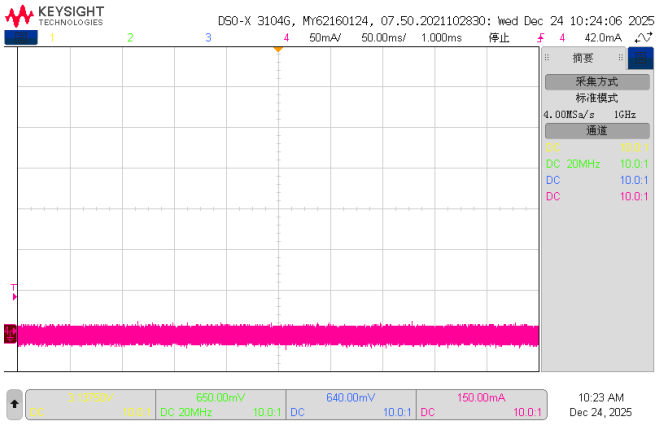


Fig.5-13-1 均衡电流@Full speed

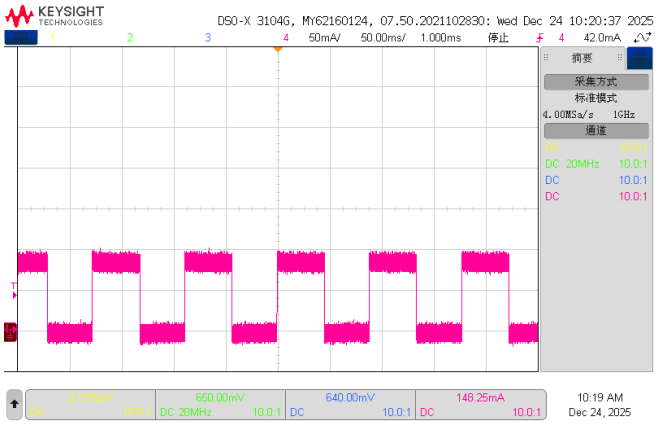


Fig.5-13-2 均衡电流@Half speed

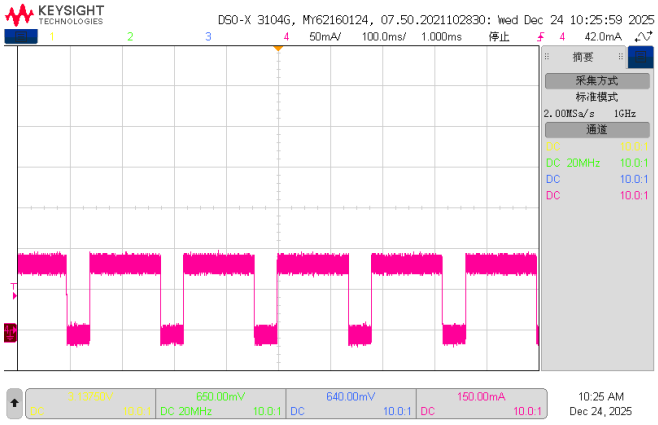


Fig.5-13-3 均衡电流@1/4 speed

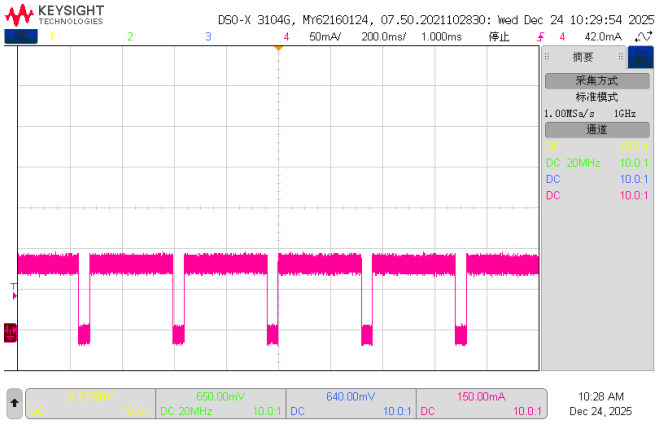


Fig.5-13-4 均衡电流@1/8 speed

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均衡电路工作时，由VC16给均衡控制电路供电，流入VC16引脚的总电流大约为12uA 乘以被均衡的电芯数量。而在测量LOOP时，会关闭均衡，故均衡期间基本不影响Cell电压采样。

Tab.5-13-2 电芯均衡时 Cell 电压采样@VCellx=4500mV

均衡状态	Cell Voltage(mV)															
	Cell1	Cell2	Cell3	Cell4	Cell5	Cell6	Cell7	Cell8	Cell9	Cell10	Cell11	Cell12	Cell13	Cell14	Cell15	Cell16
不均衡	4500	4499	4500	4499	4499	4499	4500	4499	4499	4499	4499	4499	4499	4500	4499	4499
奇均衡	4500	4499	4500	4499	4499	4499	4500	4499	4499	4499	4499	4499	4499	4500	4499	4497
偶均衡	4500	4499	4500	4499	4499	4499	4500	4499	4499	4499	4499	4499	4499	4500	4499	4498

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5.14 一次保护

本系统设计具有完整的电池电芯保护，包括：电芯过压、电芯欠压、两级过流放电、过流充电、放电短路、过热和欠温保护等。

5.14.1 电芯过压保护

本系统设计电芯过压保护设置为 4200mV，延迟时间 1s；过压保护恢复设置为 4000mV。

Tab.5-14-1 电芯过压保护及恢复

	Cell1	Cell2	Cell3	Cell4	Cell5	Cell6	Cell7	Cell8	Cell9	Cell10	Cell11	Cell12	Cell13	Cell14	Cell15	Cell16
COV(mV)	4204	4204	4204	4203	4203	4203	4203	4204	4203	4203	4203	4203	4202	4203	4203	4204
COV_Hys(mV)	4055	4055	4053	4052	4053	4052	4054	4054	4054	4054	4053	4053	4053	4053	4053	4053

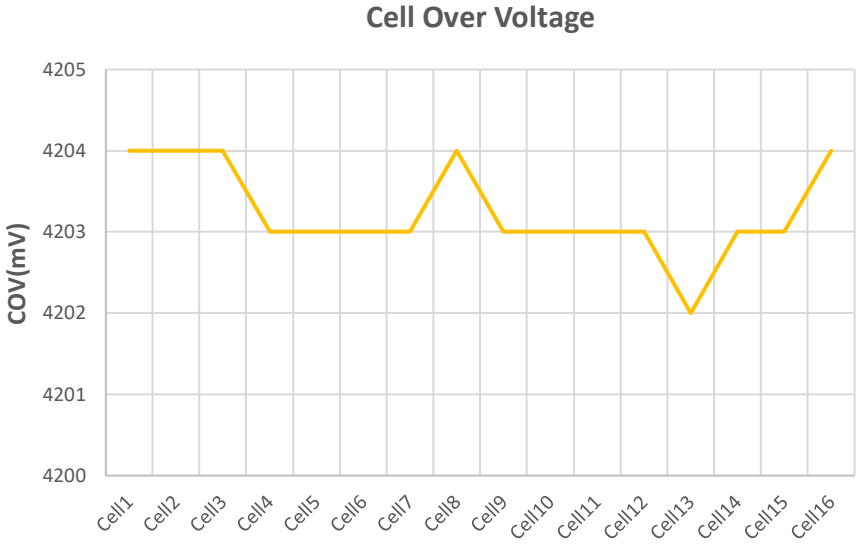
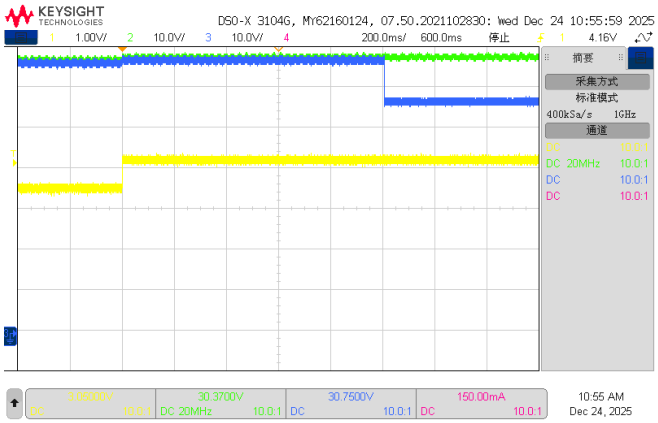
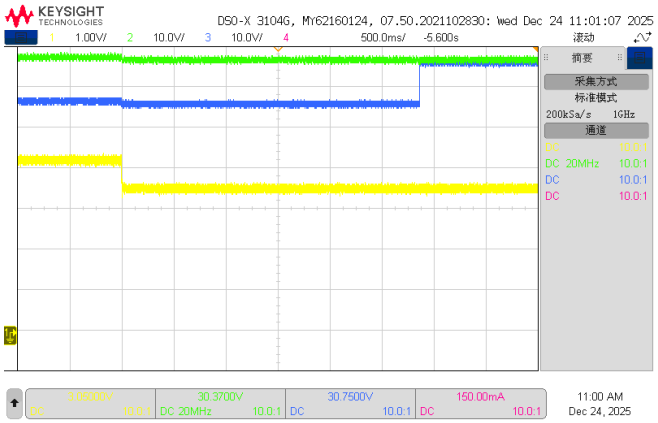


Fig.5-14-1 电芯过压保护点



CH1:Vcell1 CH2:DSG CH3:CHG

Fig.5-14-2 电芯过压保护



CH1:Vcell1 CH2:DSG CH3:CHG

Fig.5-14-3 电芯过压保护解除

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5.14.2 电芯欠压保护

本系统设计电芯欠压保护设置为 2700mV，延迟时间 1s；欠压保护恢复设置为 3000mV。

Tab.5-14-2 电芯欠压保护及恢复

	Cell1	Cell2	Cell3	Cell4	Cell5	Cell6	Cell7	Cell8	Cell9	Cell10	Cell11	Cell12	Cell13	Cell14	Cell15	Cell16
CUV(mV)	2700	2698	2696	2695	2695	2696	2696	2696	2696	2696	2697	2696	2696	2696	2696	2696
CUV_Hys(mV)	3005	3005	3005	3004	3004	3004	3003	3004	3004	3004	3005	3003	3003	3004	3004	3005

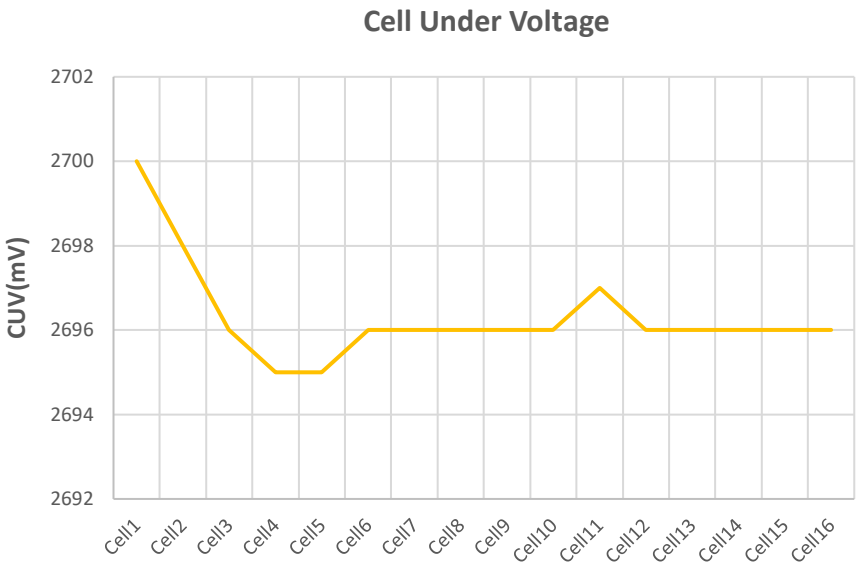
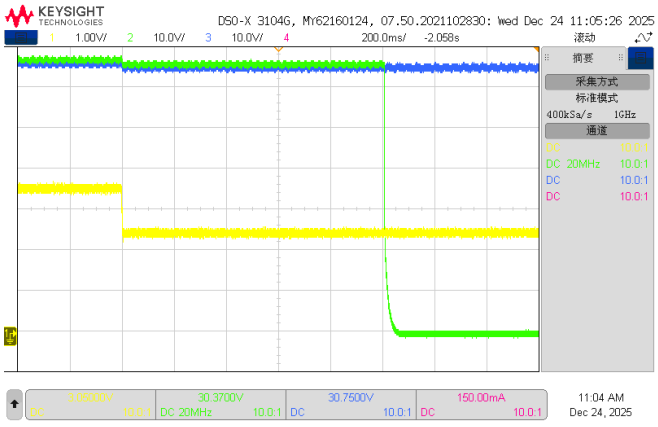
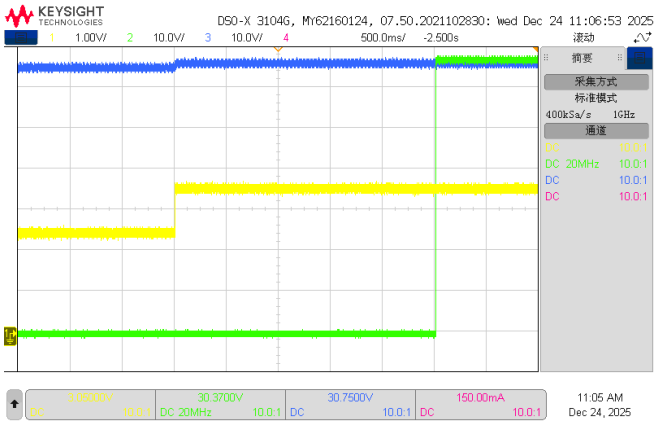


Fig.5-14-4 电芯欠压保护点



CH1:Vcell1 CH2:DSG CH3:CHG

Fig.5-14-5 电芯欠压保护



CH1:Vcell1 CH2:DSG CH3:CHG

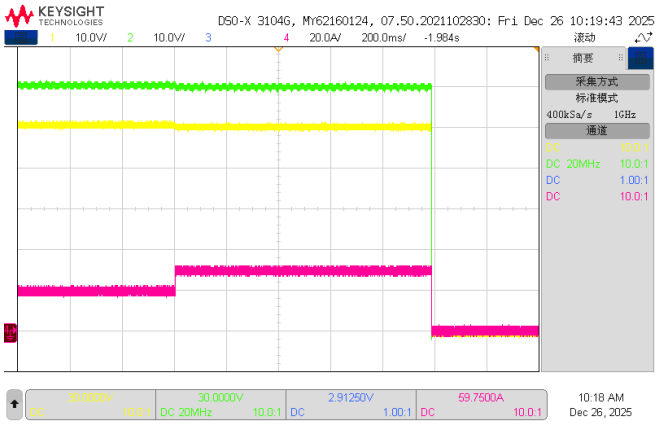
Fig.5-14-6 电芯欠压保护解除

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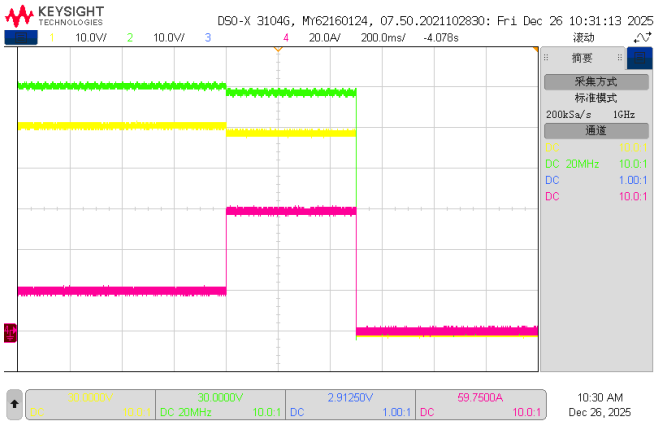
5.14.3 放电过流保护

本系统设计一级放电过流保护设置为 25A，延迟时间 1s；二级放电过流保护设置为 50A，延迟时间 0.5s。



CH1:PACK CH2:DSG CH4:ILoad

Fig.5-14-7 一级放电过流保护

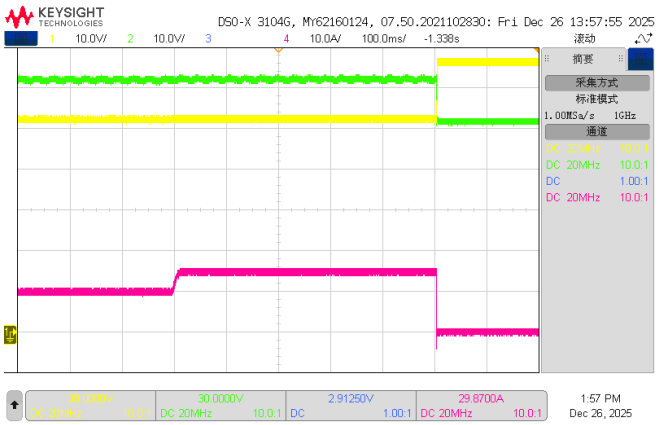


CH1:PACK CH2:DSG CH4:ILoad

Fig.5-14-8 二级放电过流保护

5.14.4 充电过流保护

本系统设计充电过流保护设置为 12A，延迟时间 0.5s。



CH1:PACK CH2:CHG CH4:ICharge

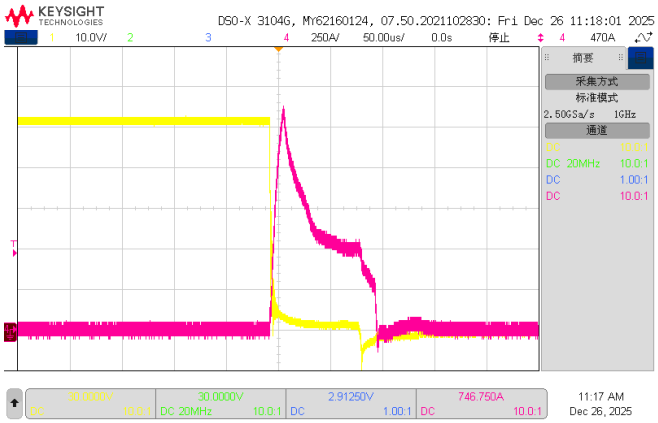
Fig.5-14-9 充电过流保护

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5.14.5 放电短路保护

本系统设计放电短路保护设置为 100A，延迟时间 60μs。



CH1:PACK CH4:ILoad

Fig.5-14-10 放电短路保护

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6 热插拔

电芯电压 4200mV，电池组总压 67.2V；充电器输出电压 72V。

Tab.6-1-1 热插拔试验

类型	插拔次数	结果
Cell 热插拔	1000	Pass
充电热插拔	1000	Pass

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7 ESD 性能

根据 IEC 61000-4-2 标准，对本系统设计进行 ESD 性能测试。EVB 带电池组上电。

Tab.7-1-1 ESD 接触放电试验

端口	结果
PACK+	±8kV
PACK-	±8kV
Cellx	±8kV

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8 历史版本

版本号	描述	日期
A	初始发布	2025 年 12 月 27
B	更新 Cell 电压采样、电流采样、温度测量精度 增加均衡时电压采样数据	2026 年 03 月 20

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